



ARS 404-21



ARS 408-21



SRR 308-21

Standard Radar Interface

Technical Documentation

ARS 404-21 (Entry)

ARS 408-21 (Premium)

SRR 308-21 (Short Range)

V 1.12

October 10, 2019

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History

Version	Date	SW Version	Change description
V 1.0	14.10.2016	ARS408 04.30.00 ARS408 04.31.00	First released version
V 1.1	18.01.2017	ARS408 04.30.00 ARS408 04.31.00	Table 36 and Table 43 – corrections
V 1.2	25.02.2017	ARS408 04.30.00 ARS408 04.31.00	Table 3 page 12
V 1.3	23.03.2017	ARS408 04.30.00 ARS408 04.31.00	Table 43 page 47 – corrections – Last service of SW 04.30.0 (Standard) and of SW 04.31.0 (Reduced Power)
V 1.4	31.05.2017	ARS408 04.30.01	Adaption for SW version 04.30.1: Radar state (0x201), Software Version Identification (0x700)
V 1.5	21.07.2017	ARS408 04.30.01	No. of clusters/objects specified <i>CollDetRegState_WarningLevel</i> regarding 0x2 changed in Table 20 Change to 196 m and 260 m regarding max. distance Change to 196 m and 1200 m regarding extended max. distance
V 1.6	01.08.2017	ARS408 04.30.01	declaration of the ambiguity (darkness, fuzziness) of the clusters on page 40, table 34
V 1.7	11.09.2017	ARS408 04.30.01 ARS404 04.10.00	Adaptions regarding ARS404 and minor clarifications
V 1.8	30.10.2018	ARS408 04.30.01 ARS404 04.10.00 SRR308 03.20.00	Additional documentation for SRR308
V 1.9	26.11.2018	ARS408 04.30.01 ARS404 04.10.00 SRR308 03.20.00	Update for SRR308
V 1.10	10.01.2019	ARS408 04.30.01 ARS404 04.10.00 SRR308 03.20.00	Minor changes
V 1.11	07.03.2019	ARS408 04.30.01 ARS404 04.10.00 SRR308 03.20.02	Pedestrian class removed
V 1.12	10.10.2019	ARS408 04.30.02 ARS404 04.10.01 SRR308 03.20.03	Minor clarifications Correction: Object_DynProp

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1. Introduction

The ARS400 / SRR300 is a Radar Sensor System developed by Continental for the Automotive Industry to realize advanced driver assistance functions.

For the ARS408 / ARS404 and SRR308, the software of the sensor was adapted to use it for general purposes applications. The adaption includes simplifications for a non-automotive usage. With the easy to use software interface it is possible to connect the sensor to a CAN network and to provide radar based environmental information to one or several evaluation units. The sensor can also be configured via CAN.

2. Operating Conditions

Please refer to *ARS 404/408-Technical Description / SRR308-Technical Description*.

3. Safety Information

Please refer to *ARS 404/408-Technical Description / SRR308-Technical Description*.

4. Interface

The ARS40X / SRR308 sensor has one CAN interface. The communication network is a CAN bus as specified in ISO 11898-2 with a transmission rate of 500 KBits/s. The sensor is not equipped with a termination resistor. A proper CAN bus termination must be realized outside the sensor.

The CAN connection of the ARS / SRR is used for configuration, sensor state output, data input and data output.

4.1. Sensor-ID Configuration

To be able to add up to eight sensors to one common CAN bus, the sensor ID can be configured. The sensor ID configuration will change the CAN message IDs of the sensor. This document describes only the CAN message IDs which are valid for a radar sensor which is configured to ID 0. For the sensor IDs 0 to 7 the CAN message ID can be calculated by the following formula:

$$\text{CAN-Msg-ID-X} = \text{CAN-Msg-ID-0} + (\text{Sensor-ID-X} * 0x10).$$

For example, the configuration CAN message 0x200 for sensor ID 0, will become 0x210 for sensor ID 1. After configuration of the sensor ID, the sensor will only react to this new CAN message ID for any configuration. Additionally, the CAN message IDs of all sensor output messages will change according to the formula above. The only exception is the relay control message (0x8), which has the same CAN ID for all sensors.

The following table gives the CAN message IDs for a sensor configured to ID 0:

Table 1: Sensor CAN messages for sensor ID 0.

In/Out	CAN ID	Message Name	Content	Section
In	0x200	RadarCfg	Radar sensor configuration	6.1
Out	0x201	RadarState	Radar status	7.1
In	0x202	FilterCfg	Cluster and Object filter configuration	6.2
Out	0x203	FilterState_Header	Filter status header	7.2
Out	0x204	FilterState_Cfg	Filter configuration status	7.3
In	0x400	CollDetCfg	Collision detection configuration	6.3
In	0x401	CollDetRegionCfg	Collision detection region configuration	6.4
Out	0x408	CollDetState	Collision detection status	7.4
Out	0x402	CollDetRegionState	Collision detection region status	7.5
In	0x300	SpeedInformation	Vehicle (sensor platform) speed	8.1
In	0x301	YawRateInformation	Vehicle (sensor platform) yaw rate	8.2
Out	0x600	Cluster_0_Status	Cluster status (list header)	9.1
Out	0x701	Cluster_1_General	Cluster general information	9.2
Out	0x702	Cluster_2_Quality	Cluster quality information	9.3
Out	0x60A	Obj_0_Status	Object status (list header)	10.1
Out	0x60B	Obj_1_General	Object general information	10.2
Out	0x60C	Obj_2_Quality	Object quality information	10.3
Out	0x60D	Obj_3_Extended	Object extended information	10.4

Out	0x60E	Obj_4_Warning	Object collision detection warnings	10.5
Out	0x700	VersionID	Software Version Identification	7.7
Out	0x8	CollDetRelayCtrl	Control message for relay	7.6

5. Description

The ARS / SRR sensor uses radar radiation to analyze its surroundings. The reflected signals are processed and after multiple steps they are available in form of clusters and objects. Clusters are radar reflections with information like position, velocity and signal strength. They are newly evaluated every cycle. In contrast to this, objects have a history and dimension. They consist of tracked clusters.

The position is given in a Cartesian coordinate system relative to the sensor as shown in Figure 1. The velocity is calculated relative to an assumed vehicle course. The course is determined by using the speed and yaw rate information assuming that the radar sensor is mounted on the front and the movement is in longitudinal direction. If speed and yaw rate information is missing, it will be set to default values: yaw rate = 0 deg/s, speed = 0 m/s and standstill.

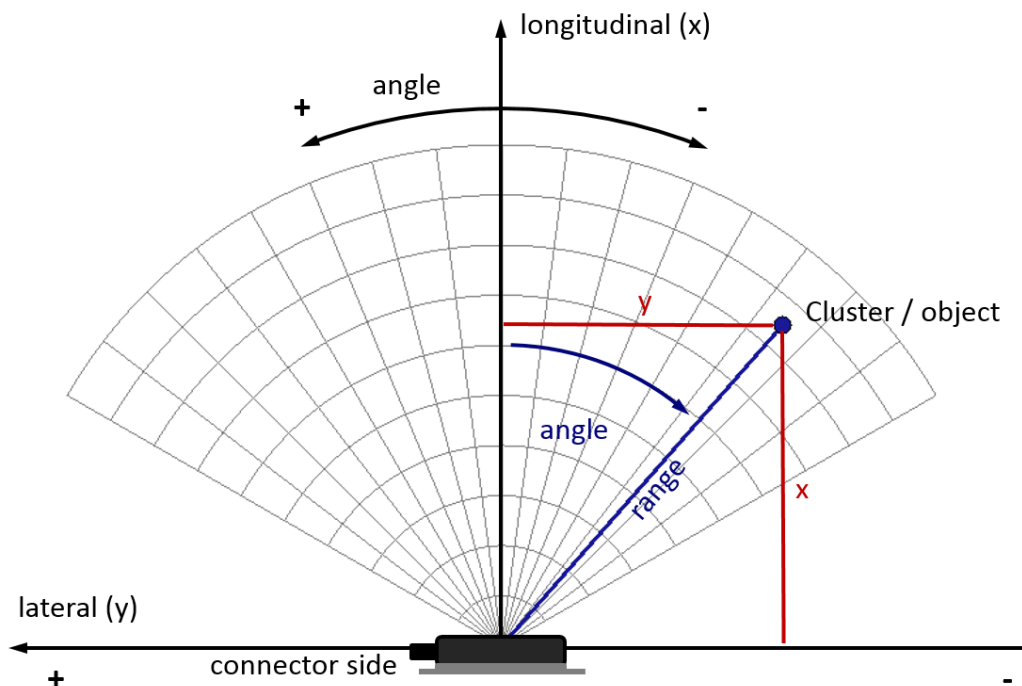


Figure 1: ARS40X coordinate system for clusters and objects.

Optionally, the output cluster and object lists can be filtered by setting filter criteria based on their attributes. Like this, the clusters or objects of interest that are sent on the CAN-bus can be selected. The filters for different attributes can be combined. More information on the filter configurations can be found in section 6.2.

Furthermore, a region-based collision detection is implemented on the sensor. If objects are detected within up to eight defined regions, the sensor will issue a warning message. More information on the collision detection configuration can be found in sections 6.3 and 6.4.

6. Configuration Messages

The basic configuration parameters of the radar sensor can be set with message **RadarCfg** (0x200). It is not necessary to cyclically repeat the configuration message. To store the configuration in the non-volatile memory (NVM) to be automatically set at startup on any subsequent power up, the signal **RadarCfg_StoreInNVM** has to be set to **active** (0x1). It is important to note that the number of transmissions to the NVM should be kept to a minimum as this could reduce the service life of the memory. A data storage to the NVM is not required in case the configuration is only used within one power cycle. Detailed descriptions of all configuration parameters can be found in section 6.1.

Additionally, filter criteria for the cluster and object list can be set by message **FilterCfg** (0x202). The kind of filter to configure is set by multiplexor signals **FilterCfg_Type** (filter clusters or objects) and **FilterCfg_Index** (filter criterion). More details can be found in section 6.2.

The region-based collision detection can be configured with messages **CollDetCfg** (0x400) and **CollDetRegCfg** (0x401) for up to eight different regions. Details can be found in sections 6.3 and 6.4.

The filter and collision detection configurations are always stored in Nvm (when the configuration has been changed) and set automatically when the sensor restarts.

6.1. Radar configuration (0x200)

The ARS / SRR can be configured via message **RadarCfg** (0x200). The parameters can be changed individually or in combinations. For each parameter, the message contains a validity bit (e.g. **RadarCfg_MaxDistance_valid** for the configuration parameter **RadarCfg_MaxDistance**). Only if the validity bit is set to **valid** (0x1), the corresponding configuration will be updated in the ARS, otherwise it is ignored.

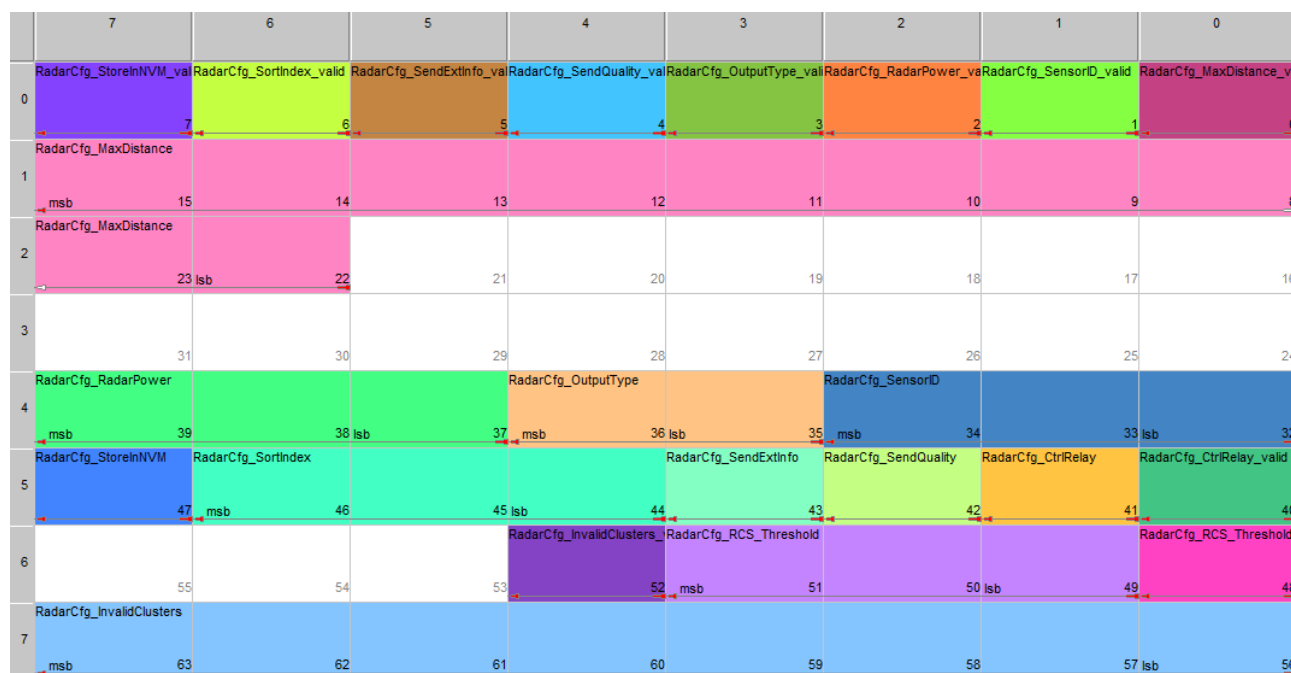


Figure 2: RadarCfg- message layout (0x200).

Table 2: RadarCfg- message content (0x200).

Signal	Start	Len	Min	Max	Res	Unit
RadarCfg_MaxDistance_valid	0	1	0	1	1	0x0: invalid 0x1: valid
RadarCfg_SensorID_valid	1	1	0	1	1	0x0: invalid 0x1: valid
RadarCfg_RadarPower_valid	2	1	0	1	1	0x0: invalid 0x1: valid
RadarCfg_OutputType_valid	3	1	0	1	1	0x0: invalid 0x1: valid
RadarCfg_SendQuality_valid	4	1	0	1	1	0x0: invalid 0x1: valid
RadarCfg_SendExtInfo_valid	5	1	0	1	1	0x0: invalid 0x1: valid
RadarCfg_SortIndex_valid	6	1	0	1	1	0x0: invalid 0x1: valid
RadarCfg_StoreInNVM_valid	7	1	0	1	1	0x0: invalid 0x1: valid
RadarCfg_MaxDistance	22	10	0	2046	2	m
RadarCfg_SensorID	32	3	0	7	1	Sensor ID 0x0 to 0x7
RadarCfg_OutputType	35	2	0	3	1	0x0: none 0x1: send objects 0x2: send clusters
RadarCfg_RadarPower ¹	37	3	0	7	1	0x0: Standard 0x1: -3dB Tx gain 0x2: -6dB Tx gain 0x3: -9dB Tx gain
RadarCfg_CtrlRelay_valid	40	1	0	1	1	0x0: invalid 0x1: valid
RadarCfg_CtrlRelay	41	1	0	1	1	0x0: inactive 0x1: active
RadarCfg_SendQuality	42	1	0	1	1	0x0: inactive 0x1: active
RadarCfg_SendExtInfo	43	1	0	1	1	0x0: inactive 0x1: active
RadarCfg_SortIndex	44	3	0	7	1	0x0: no sorting 0x1: sorted by range 0x2: sorted by RCS
RadarCfg_StoreInNVM	47	1	0	1	1	0x0: inactive 0x1: active

¹ Not applicable for SRR308, any change in value of this signal is ignored.

RadarCfg_RCS_Threshold_valid	48	1	0	1	1	0x0: invalid 0x1: valid
RadarCfg_RCS_Threshold	49	3	0	7	1	0x0: Standard 0x1: High sensitivity
RadarCfg_InvalidClusters_valid ²	52	1	0	1	1	0x0: invalid 0x1: valid
RadarCfg_InvalidClusters ^{2,3}	56	8	0	255	1	0x00: Disable invalid cluster 0x01: Enable all invalid clusters 0x02 Enable low RCS dynamic 0x04 Enable low RCS static 0x08 Enable invalid range rate 0x10 Enable range < 1m 0x20 Enable ego mirror 0x40 Enable wrapped stationary

Table 3: RadarCfg – signal description (0x200).

Start	Signal	Description
32	RadarCfg_SensorID	Sensor ID 0 – 7 (see also section 4.1 for further explanation)
1	RadarCfg_SensorID_valid	Allow change of sensor ID if true

² Available only for SRR308³ Any valid combination other than the mentioned values will enable all the respective invalid clusters as per Cluster_InvalidReasonBitfield signal

22	RadarCfg_MaxDistance	Maximum detection distance (ARS408: value for far scan - near scan maximum distance is set to 1/2 of this value). The maximum distance will also change the range resolution proportionally. For example, 200 m maximum distance will lead to: - far scan 200 m with 1.79 m range resolution, - near scan 100 m with 0.42 m range resolution. ARS408: Standard Range Version: 196 – 260 m Extended Range Version: 196 – 1200 m ARS404: Standard Range Version: 150 – 190 m Extended Range Version: 90 – 1000 m SRR308: Standard Range Version: 96 m Extended Range Version: 96 - 192 m
0	RadarCfg_MaxDistance_valid	Allow change of maximum distance if true
37	RadarCfg_RadarPower	Configures the transmitted radar power (Tx attenuation). The output RCS of cluster and objects will be compensated for this attenuation. Reducing the output power can improve detection in case of close distance scenarios or inside rooms.
2	RadarCfg_RadarPower_valid	Allow change of radar output power if true
35	RadarCfg_OutputType	Configures the data output to clusters (0x2) or objects (0x1)
3	RadarCfg_OutputType_valid	Allow change of output type if true
42	RadarCfg_SendQuality	Cluster or object quality information (message 0x60C or 0x702) is sent if true
4	RadarCfg_SendQuality_valid	Allow change of quality message option if true
43	RadarCfg_SendExtInfo	Extended information (message 0x60D) is sent for objects if true (if clusters are selected as output type this value is ignored)
5	RadarCfg_SendExtInfo_valid	Allow change of extended info message option if true
44	RadarCfg_SortIndex	Selects the sorting index for the object list (ignored for clusters as they are always sorted by range)
6	RadarCfg_SortIndex_valid	Allow change of sorting index if true
41	RadarCfg_CtrlRelay	Relay control message (0x8) is sent if true and the collision detection is activated
40	RadarCfg_CtrlRelay_valid	Allow change of relay control output if true
49	RadarCfg_RCS_Threshold	Sets the sensitivity of the cluster detection to standard (0x0) or high sensitivity (0x1)
48	RadarCfg_RCS_Threshold_valid	Allow change of RCS threshold option if true

47	RadarCfg_StoreInNVM	Stores the current configuration to non-volatile memory to be read and set at sensor startup.
7	RadarCfg_StoreInNVM_valid	Allow storing to non-volatile memory if true
53	RadarCfg_InvalidClusters ⁴	SRR308 only: Enables selection of invalid clusters to be output in the cluster list. This value is considered as a bitfield according to the cluster invalid reason.
52	RadarCfg_InvalidClusters_valid ⁴	Allows changing the invalid clusters parameter if true

6.1.1 Maximum number of CAN messages

With the RadarCfg message it is configured which messages will be send out by the radar sensor. Within one radar cycle a maximum number of approximately 250 CAN messages can be sent out. In case there are too many objects or cluster detected by the radar and too many output messages are configured, the radar will not send out all expected messages. To prevent this situation either the configured output messages must be reduced, or the maximum number of objects/ clusters must be limited by a filter configuration.

6.2. Cluster and Object filter configuration (0x202)

With the multiplex message **FilterCfg** (0x202), output filters for the cluster or object list can be configured. The kind of filter to be configure is set by multiplexor signals

FilterCfg_Type (filter clusters or objects) and **FilterCfg_Index** (filter criteria). In Table 4, the possible filter indices are described.

The filters are designed as pass though filters, i.e. for each filter criterion, a valid range with minimum (**FilterCfg_Min_xxx**) and maximum (**FilterCfg_Max_xxx**) value can be specified. Exception is the filter index 0x0 (NofObj) for which only a maximum value can be set.

When a filter configuration message is sent, the sensor replies with the messages **FilterState_Header** (0x203) with the number of configured filters and one message **FilterState_Cfg** (0x204) for the filter that has been changed.

Table 4: Possible filter criteria selectable with FilterCfg_Index. The columns **C** and **O** specify if this filter can be applied to **clusters** or **objects**, respectively.

Index	Filter criterium	C	O	Description
0x0	NofObj	x	x	Limits the maximum number of cluster or objects to send out (minimum value is ignored)
0x1	Distance	x	x	Radial distance in m [$r = \sqrt{x^2 + y^2}$]
0x2	Azimuth	x	x	Azimuth angle in degree [$a = \arctan(y/x)$]
0x3	VrelOncome	x	x	Radial velocity in sensor line-of-sight in m/sec of oncoming clusters or objects (all departing clusters and objects are ok)

⁴ Available only for SRR308

0x4	VrelDepart	x	x	Radial velocity in sensor line-of-sight in m/sec of departing clusters or objects (all oncoming clusters and objects are ok)
0x5	RCS	x	x	RCS value (Radar cross section) in dBm ²
0x6	Lifetime		x	Life time (since first detection) in seconds
0x7	Size		x	Object size as area in m ² (length x width)
0x8	ProbExists		x	Probability of existence, i.e. probability for being a real target and not a sensor artifact caused by multipath etc. Possible values are - 0x0: 0% - 0x1: 25% - 0x2: 50% - 0x3: 75% - 0x4: 90% - 0x5: 99% - 0x6: 99.9% - 0x7: 100%
0x9	Y		x	Y-position in m (lateral distance)
0xA	X		x	X-position in m (longitudinal distance)
0xB	VYRightLeft		x	Lateral velocity component in m/sec for right-left moving objects (all left-right moving objects are ok)
0xC	VXOncome		x	Longitudinal velocity component in m/sec for oncoming objects (all departing objects are ok)
0xD	VYLeftRight		x	Lateral velocity component in m/sec for left-right moving objects (all right-left moving objects are ok)
0xE	VXDepart		x	Longitudinal velocity component in m/sec for departing objects (all oncoming objects are ok)
0xF	Class		x	Filters objects having a specific classification which is output in the signal Object_Class in the message Object_3_Extended. Max value is used as a bitfield for the configuration. Bit 0-7 are corresponding to the object classification 0-7. Min value is ignored.

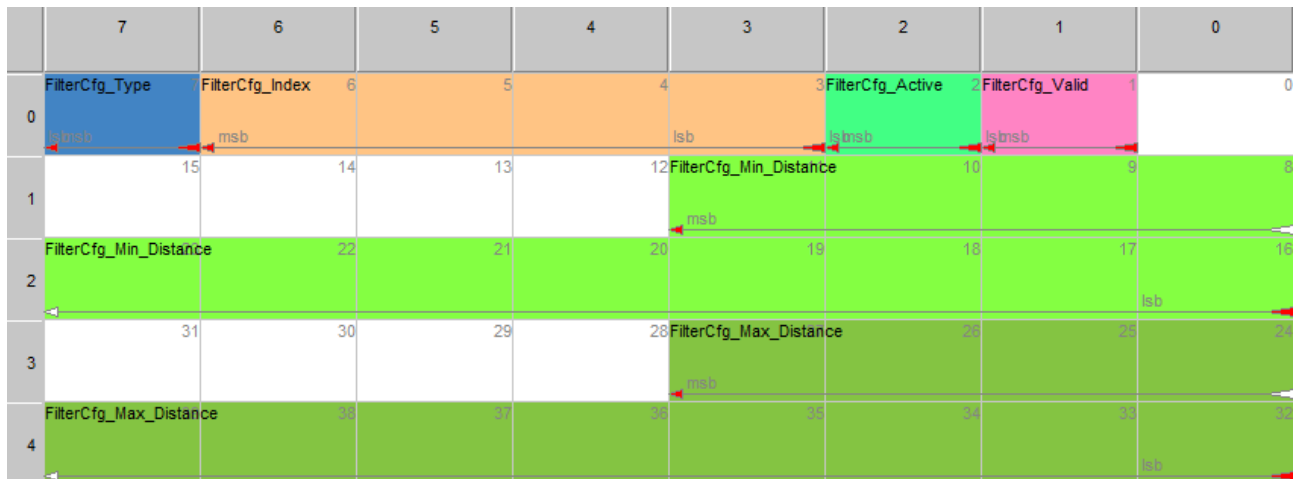


Figure 3: FilterCfg - message layout for FilterCfg_Index=0x1 (0x202) with 12 bit length for minimum and maximum value.

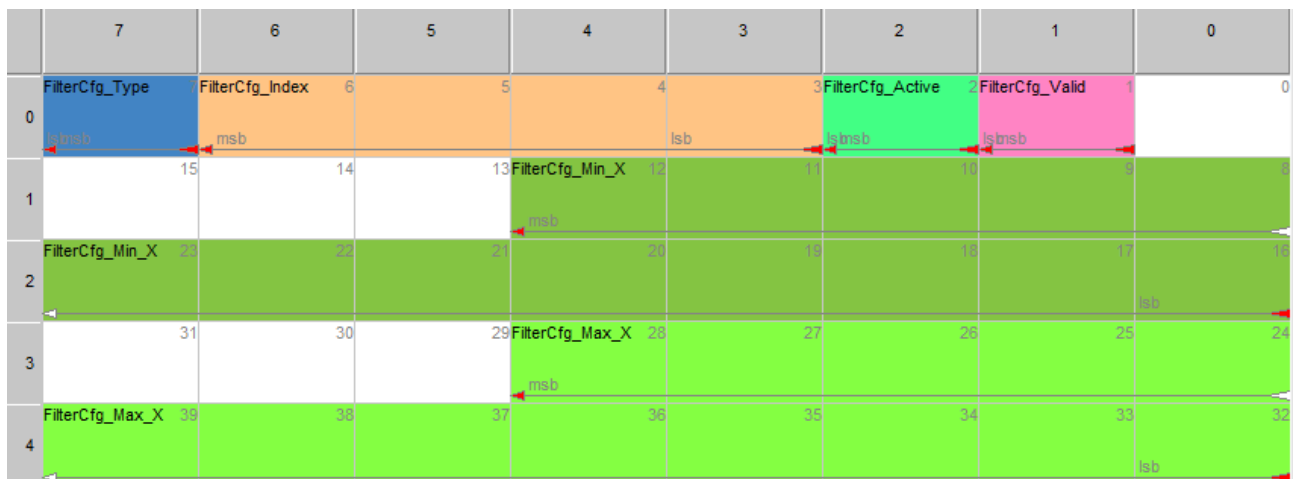


Figure 4: FilterCfg - message layout for FilterCfg_Index=0xA (0x202) with 13 bit length for minimum and maximum value.

Table 5: FilterCfg - message content (0x202).

Signal	Start	Len	Min	Max	Res	Unit
FilterCfg_Valid	1	1	0	1	1	0x0: invalid 0x1: valid
FilterCfg_Active	2	1	0	1	1	0x0: inactive 0x1: active
FilterCfg_Index	3	4	0	15	1	see Table 4
FilterCfg_Type	7	1	0	1	1	0x0: Cluster filter 0x1: Object filter
FilterCfg_Min_NoObj	16	12	0	4095	1	ignored
FilterCfg_Min_Distance	16	12	0	409.5	0.1	m
FilterCfg_Min_Azimuth	16	12	-50	52.375	0.025	deg

FilterCfg_Min_VrelOncome	16	12	0	128.993	0.0315	m/s
FilterCfg_Min_VrelDepart	16	12	0	128.993	0.0315	m/s
FilterCfg_Min_RCS	16	12	-50	52.375	0.025	dBm ²
FilterCfg_Min_Lifetime	16	12	0	409.5	0.1	s
FilterCfg_Min_Size	16	12	0	102.375	0.025	m ²
FilterCfg_Min_ProbExists	16	12	0	7	1	0x0: 0% 0x1: 25% 0x2: 50% 0x3: 75% 0x4: 90% 0x5: 99% 0x6: 99.9% 0x7: 100%
FilterCfg_Min_Y	16	12	-409.5	409.5	0.2	m
FilterCfg_Min_X	16	13	-500	1138.2	0.2	m
FilterCfg_Min_VYRightLeft	16	12	0	128.993	0.0315	m/s
FilterCfg_Min_VXOncome	16	12	0	128.993	0.0315	m/s
FilterCfg_Min_VYLeftRight	16	12	0	128.993	0.0315	m/s
FilterCfg_Min_VXDepart	16	12	0	128.993	0.0315	m/s
FilterCfg_Min_Class	16	12	0	4095	1	ignored
FilterCfg_Max_NofObj	32	12	0	4095	1	Objects
FilterCfg_Max_Distance	32	12	0	409.5	0.1	m
FilterCfg_Max_Azimuth	32	12	-50	52.375	0.025	deg
FilterCfg_Max_VrelOncome	32	12	0	128.993	0.0315	m/s
FilterCfg_Max_VrelDepart	32	12	0	128.993	0.0315	m/s
FilterCfg_Max_RCS	32	12	-50	52.375	0.025	dBm ²
FilterCfg_Max_Lifetime	32	12	0	409.5	0.1	s
FilterCfg_Max_Size	32	12	0	102.375	0.025	m ²
FilterCfg_Max_ProbExists	32	12	0	7	1	0x0: 0% 0x1: 25% 0x2: 50% 0x3: 75% 0x4: 90% 0x5: 99% 0x6: 99.9% 0x7: 100%
FilterCfg_Max_Y	32	12	-409.5	409.5	0.2	m
FilterCfg_Max_X	32	13	-500	1138.2	0.2	m
FilterCfg_Max_VYRightLeft	32	12	0	128.993	0.0315	m/s
FilterCfg_Max_VXOncome	32	12	0	128.993	0.0315	m/s

FilterCfg_Max_VYLeftRight	32	12	0	128.993	0.0315	m/s
FilterCfg_Max_VXDepart	32	12	0	128.993	0.0315	m/s
FilterCfg_Max_Class	32	12	0	4095	1	Bitfield

Table 6: FilterCfg - signal description (0x202).

Start	Signal	Description
1	FilterCfg_Valid	Allow change of filter configuration if true
2	FilterCfg_Active	De-/activate filter configuration for specified filter criterion (FilterCfg_Index) and type (FilterCfg_Type)
3	FilterCfg_Index	Multiplexor to specify which filter criterion to configure
7	FilterCfg_Type	Multiplexor to choose between object or cluster filter configuration
16	FilterCfg_Min_xxx	Minimum value for filter criterion. The content depends on FilterCfg_Index (see Table 4 for more details)
32	FilterCfg_Max_xxx	Maximum value for filter criterion. The content depends on FilterCfg_Index (see Table 4 for more details)

6.3. Collision detection configuration (0x400)

Region-based collision detection can be activated with message **CollDetCfg** (0x400). With activated collision detection, the sensor sends cyclically (once per second) the message **CollDetState** (0x408) and messages **CollDetRegionState** (0x402) for all configured regions.

Collision detection is only possible with objects, if **RadarCfg_OutputType** is set to **send Objects** (0x2). Then, the message **Obj_4_Warning** (0x60E) is send for each object, indicating if the object is violating any regions.

Furthermore, the relay control message **CollDetRelayCtrl** (0x8) is sent, if it is activated with **RadarCfg_CtrlRelay**. It contains a bit field with the current warning state of the regions, to control a relay.

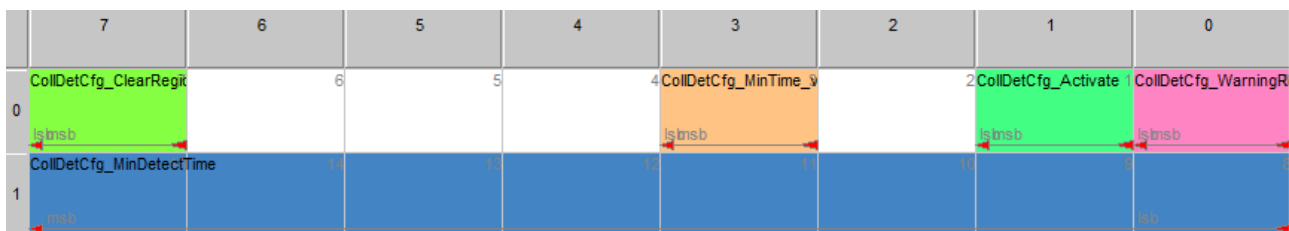


Figure 5: CollDetCfg - message layout (0x400).

Table 7: CollDetCfg - message content (0x400).

Signal	Start	Len	Min	Max	Res	Unit
CollDetCfg_WarningReset	0	1	0	1	1	0x0: idle 0x1: reset warnings

CollDetCfg_Activation	1	1	0	1	1	0x0: inactive 0x1: active
CollDetCfg_MinTime_valid	3	1	0	1	1	0x0: invalid 0x1: valid
CollDetCfg_ClearRegions	7	1	0	1	1	0x0: idle 0x1: clear regions
CollDetCfg_MinTime	8	8	0.0	25.5	0.1	sec

Table 8: CollDetCfg - signal description (0x400).

Start	Signal	Description
0	CollDetCfg_WarningReset	Reset currently active warnings of all regions (CollDetRegState_WarningLevel) if true
1	CollDetCfg_Activation	De-/activate collision detection function
3	CollDetCfg_MinTime_valid	Allow change of time parameter if true
7	CollDetCfg_ClearRegions	Clear all region configurations (set all to inactive) if true
8	CollDetCfg_MinTime	Minimum time an object needs to be detected inside the region before a warning is triggered

6.4. Collision detection region configuration (0x401)

Eight different rectangular regions can be configured with message **CollDetRegCfg** (0x401). Each region is defined by two points with x and y coordinates spanning a rectangle as shown in Figure 6. Collision regions can also overlap.

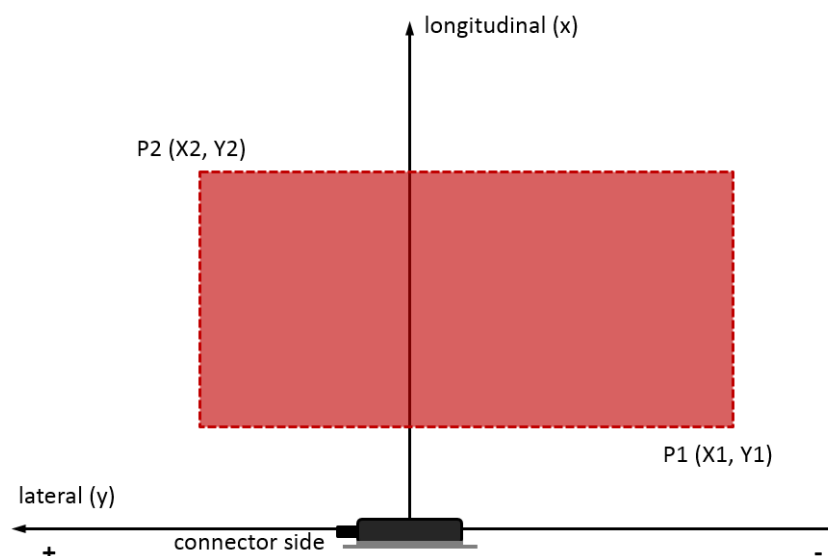


Figure 6: Coordinates for defining a collision detection region.

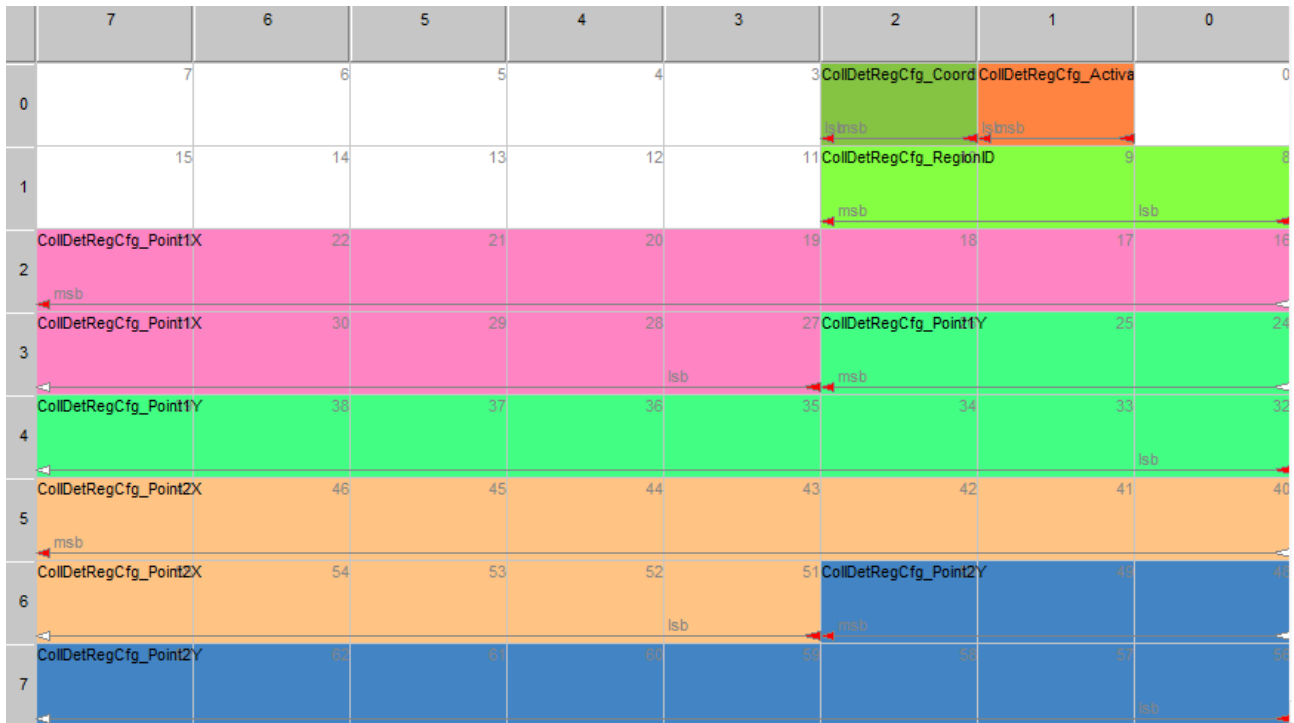


Figure 7: CollDetRegionCfg - message layout (0x401).

Table 9: CollDetRegionCfg - message content (0x401).

Signal	Start	Len	Min	Max	Res	Unit
CollDetRegCfg_Activation	1	1	0	1	1	0x0: inactive 0x1: active
CollDetRegCfg_Coordinates_valid	2	1	0	0	1	0x0: invalid 0x1: valid
CollDetRegCfg_RegionID	8	3	0	7	1	-
CollDetRegCfg_Point1X	27	13	-500	1138.2	0.2	m
CollDetRegCfg_Point1Y	32	11	-204.6	204.8	0.2	m
CollDetRegCfg_Point2X	51	13	-500	1138.2	0.2	m
CollDetRegCfg_Point2Y	56	11	-204.6	204.8	0.2	m

Table 10: CollDetRegionCfg - signal description (0x401).

Start	Signal	Description
1	CollDetRegCfg_Activation	De-/activate current region
2	CollDetRegCfg_Coordinates_valid	Allow change of current region's coordinates if true
5	CollDetRegCfg_RegionID	ID of current region to configure (0-7)
27	CollDetRegCfg_Point1X	1 st X (longitudinal) coordinate of rectangular region
32	CollDetRegCfg_Point1Y	1 st Y (lateral) coordinate of rectangular region
51	CollDetRegCfg_Point2X	2 nd X (longitudinal) coordinate of rectangular region
56	CollDetRegCfg_Point2Y	2 nd Y (lateral) coordinate of rectangular region

7. State Output

The sensor is always sending cyclically (once per second) the current configuration and sensor state in message **RadarState** (0x201) and current firmware version in message **VersionID** (0x700).

For the filter configuration, the sensor only sends the current configuration as a reply to the filter configuration message **FilterCfg** (0x202). The sensor replies with the message **FilterState_Header** (0x203) with the number of configured filters and one message **FilterState_Cfg** (0x204) for the filter that has been configured.

When collision detection is active, additionally, the sensor sends the current collision detection configuration and warning state in message **CollDetState** (0x408) and for the individual regions in messages **CollDetRegionState** (0x402). Furthermore, the relay control message **CollDetRelayCtrl** (0x8) is sent, if it is activated with **RadarCfg_CtrlRelay**. They are also sent cyclically (once per second).

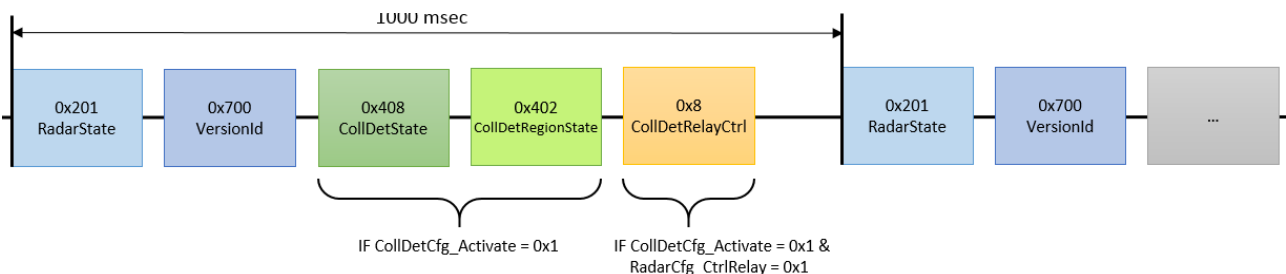


Figure 8: Overview of state messages that are sent cyclically by the radar.

7.1. Radar state (0x201)

The message **RadarState** (0x201) is sent by the sensor in a regular interval (once per second). After configuring a radar configuration parameter, the corresponding signal in message 0x201 can be checked in order to verify that the configuration change has been accepted.

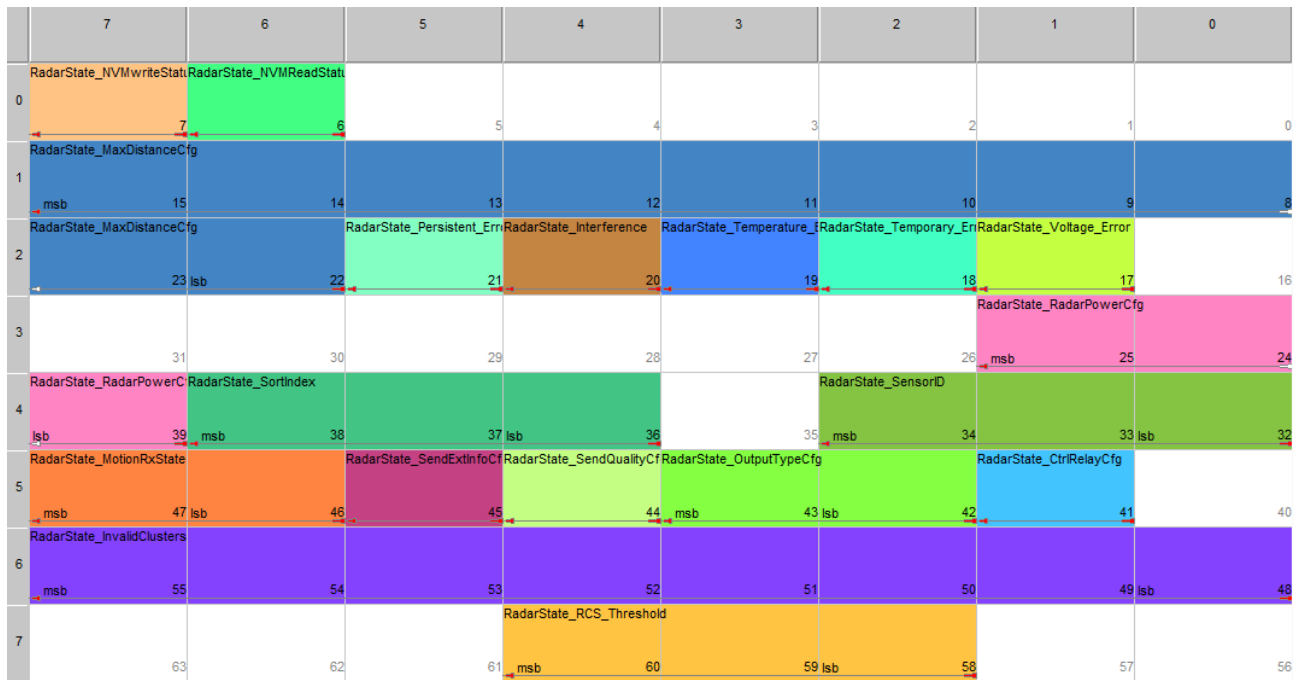


Figure 9: RadarState - message layout (0x201).

Table 11: RadarState - message content (0x201).

Signal	Start	Len	Min	Max	Res	Unit
RadarState_NVMReadStatus	6	1	0	1	1	0x0: failed 0x1: successful (default)
RadarState_NVMwriteStatus	7	1	0	1	1	0x0: failed (default) 0x1: successful
RadarState_MaxDistanceCfg	22	10	0	2046	2	m (196 m - 260 m are supported in the standard version) (default: 250 m)
RadarState_Persistent_Error	21	1	0	1	1	0x0: No error (default) 0x1: Persistent error active
RadarState_Interference	20	1	0	1	1	0x0: No interference (default) 0x1: Interference detected
RadarState_Temperature_Error	19	1	0	1	1	0x0: No error (default) 0x1: Temperature error active
RadarState_Temporary_Error	18	1	0	1	1	0x0: No error (default) 0x1: Temporary error active
RadarState_Voltage_Error	17	1	0	1	1	0x0: No error (default) 0x1: Voltage error active
RadarState_SensorID	32	3	0	7	1	Current sensor ID (0-7) (default: 0)

RadarState_SortIndex	36	3	0	7	1	0x0: no sorting (default) 0x1: sorted by range 0x2: sorted by RCS
RadarState_RadarPowerCfg	39	3	0	7	1	0x0: Standard (default) 0x1: -3dB Tx gain 0x2: -6dB Tx gain 0x3: -9dB Tx gain
RadarState_CtrlRelayCfg	41	1	0	1	1	0x0: inactive (default) 0x1: active
RadarState_OutputTypeCfg	42	2	0	3	1	0x0: none 0x1: send objects 0x2: send clusters (default)
RadarState_SendQualityCfg	44	1	0	1	1	0x0: inactive (default) 0x1: active
RadarState_SendExtInfoCfg	45	1	0	1	1	0x0: inactive (default) 0x1: active
RadarState_MotionRxState	46	2	0	3	1	0x0: input ok 0x1: speed missing 0x2: yaw rate missing 0x3: speed and yaw rate missing (default)
RadarState_RCS_Threshold	58	3	0	7	1	0x0 Standard (default) 0x1 High sensitivity
RadarState_InvalidClusters ⁵	48	8	0	255	1	0x0 All invalid clusters disabled (default) 0x1 All invalid clusters enabled 0x2 Low RCS dynamic enabled 0x4 Low RCS static enabled 0x8 Invalid range rate enabled 0x10 Range <1m enabled 0x20 Ego mirror enabled 0x40 Wrapped stationary enabled

⁵ Available only for SRR308

Table 12: RadarState - signal description (0x201).

Start	Signal	Description
6	RadarState_NVMReadStatus	State of reading the configuration parameters from non-volatile memory at startup
7	RadarState_NVMwriteStatus	State of storing a configuration parameter to non-volatile memory (initially this value is set to 0x0 and set to 0x1 after a configuration has been sent and successfully stored)
17	RadarState_Voltage_Error	Error will be active if the operating voltage is below or above the defined range for more than 5 seconds.
18	RadarState_Temporary_Error	A temporary error which will most probably disappear after a sensor reset has been detected.
19	RadarState_Temperature_Error	Error will be active if the temperature is below or above the defined range.
20	RadarState_Interference	Interference with another radar sensor has been detected.
21	RadarState_Persistent_Error	An internal error which might not disappear after a reset has been detected.
22	RadarState_MaxDistanceCfg	Current configuration for maximum distance of far scan
32	RadarState_SensorID	Sensor ID 0 - 7
36	RadarState_SortIndex	Current configuration of sorting index for object list
39	RadarState_RadarPowerCfg	Current configuration of transmitted radar power parameter
41	RadarState_CtrlRelayCfg	True if relay control message is sent
42	RadarState_OutputTypeCfg	Currently selected output type as either clusters or objects
44	RadarState_SendQualityCfg	True if quality information is sent for clusters or objects
45	RadarState_SendExtInfoCfg	True if extended information is sent for objects
46	RadarState_MotionRxState	Shows the state of the speed and yaw rate input signals
58	RadarState_RCS_Threshold	If true, the sensor's high sensitivity mode is active
61	RadarState_InvalidClusters ⁶	Shows which of the invalid clusters are selected for output.

⁶ Available only for SRR308

7.2. Cluster and Object filter configuration state header (0x203)

After configuring a filter parameter, the sensor replies with messages **FilterState_Header** (0x203) and filter status message **FilterState_Cfg** (0x204) of the filter that was modified.

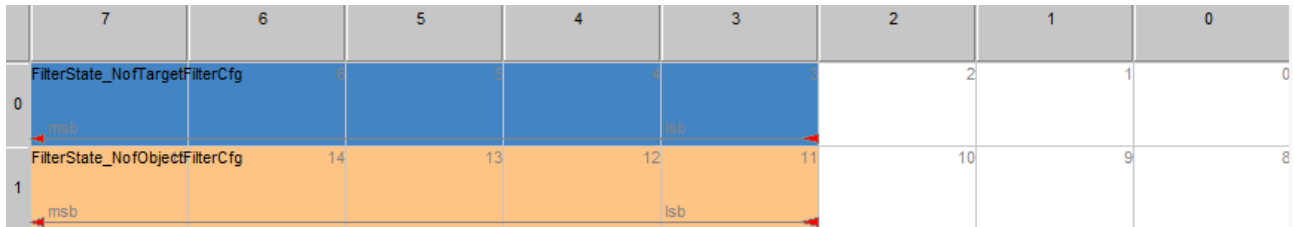


Figure 10: FilterState_Header - message layout (0x203).

Table 13: FilterState_Header - message content (0x203).

Signal	Start	Len	Min	Max	Res	Unit
FilterState_NofClusterFilterCfg	3	5	0	31	1	-
FilterState_NofObjectFilterCfg	11	5	0	31	1	-

Table 14: FilterState_Header - signal description (0x203).

Start	Signal	Description
3	FilterState_NofClusterFilterCfg	Number of filter status messages for configured Cluster filter
11	FilterState_NofObjectFilterCfg	Number of filter status messages for configured Object filter

7.3. Cluster and Object filter configuration state (0x204)

After configuring a filter parameter, the sensor replies with messages **FilterState_Header** (0x203) and filter status message **FilterState_Cfg** (0x204) of the filter that was modified.

The kind of filter is defined by multiplexor signals **FilterState_Type** and **FilterState_Index**. In Table 15, the possible filter indices are described. The table is a copy of Table 4 and just repeated here for convenience.

Table 15: Possible filter criteria selectable with FilterState_Index. The columns C and O specify if this filter can be applied to clusters or objects, respectively.

Index	Filter criterium	C	O	Description
0x0	NofObj	x	x	Limits the maximum number of cluster or objects to send out (minimum value is ignored)
0x1	Distance	x	x	Radial distance in m [$r = \sqrt{x^2 + y^2}$]
0x2	Azimuth	x	x	Azimuth angle in degree [$a = \text{atan}(y/x)$]
0x3	VrelOncome	x	x	Radial velocity in sensor line-of-sight in m/sec of oncoming clusters or objects (all departing clusters and objects are ok)
0x4	VrelDepart	x	x	Radial velocity in sensor line-of-sight in m/sec of departing clusters or objects (all oncoming clusters and objects are ok)
0x5	RCS	x	x	RCS value (Radar cross section) in dBm ²
0x6	Lifetime		x	Life time (since first detection) in seconds
0x7	Size		x	Object size as area in m ² (length x width)
0x8	ProbExists		x	Probability of existence, i.e. probability for being a real target and not a sensor artefact caused by multipath etc. Possible values can be - 0x0: 0% - 0x1: 25% - 0x2: 50% - 0x3: 75% - 0x4: 90% - 0x5: 99% - 0x6: 99.9% - 0x7: 100%
0x9	Y		x	Y-position in m (lateral distance)
0xA	X		x	X-position in m (longitudinal distance)
0xB	VYRightLeft		x	Lateral velocity component in m/sec for right-left moving objects (all left-right moving objects are ok)
0xC	VXOncome		x	Longitudinal velocity component in m/sec for oncoming objects (all departing objects are ok)
0xD	VYLeftRight		x	Lateral velocity component in m/sec for left-right moving objects (all right-left moving objects are ok)
0xE	VXDepart		x	Longitudinal velocity component in m/sec for departing objects (all oncoming objects are ok)

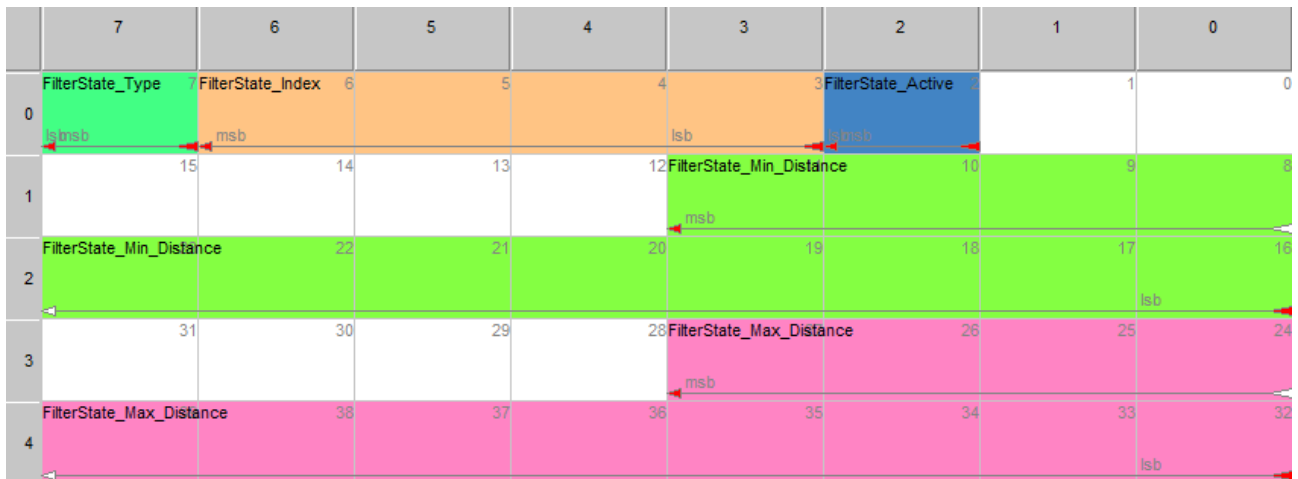


Figure 11: FilterState_Cfg - message layout for FilterState_Index=0x1 (0x204) with 12 bit length for minimum and maximum value.



Figure 12: FilterState_Cfg - message layout for FilterState_Index=0xA (0x204) with 13 bit length for minimum and maximum value.

Table 16: FilterState_Cfg - message content (0x204).

Signal	Start	Len	Min	Max	Res	Unit
FilterState_Active	2	1	0	1	1	0x0: invalid 0x1: valid
FilterState_Index	3	4	0	15	1	see Table 4
FilterState_Type	7	1	0	1	1	0x0: Cluster filter 0x1: Object filter
FilterState_Min_NofObj	16	12	0	4095	1	not used
FilterState_Min_Distance	16	12	0	409.5	0.1	m
FilterState_Min_Azimuth	16	12	-50	52.375	0.025	deg
FilterState_Min_VrelOncome	16	12	0	128.993	0.0315	m/s
FilterState_Min_VrelDepart	16	12	0	128.993	0.0315	m/s

FilterState_Min_RCS	16	12	-50	52.375	0.025	dBm ²
FilterState_Min_Lifetime	16	12	0	409.5	0.1	s
FilterState_Min_Size	16	12	0	102.375	0.025	m ²
FilterState_Min_ProbExists	16	12	0	7	1	0x0: 0% 0x1: 25% 0x2: 50% 0x3: 75% 0x4: 90% 0x5: 99% 0x6: 99.9% 0x7: 100%
FilterState_Min_Y	16	12	-409.5	409.5	0.2	m
FilterState_Min_X	16	13	-500	1138.2	0.2	m
FilterState_Min_VYLeftRight	16	12	0	128.993	0.0315	m/s
FilterState_Min_VXOncome	16	12	0	128.993	0.0315	m/s
FilterState_Min_VYRightLeft	16	12	0	128.993	0.0315	m/s
FilterState_Min_VXDepart	16	12	0	128.993	0.0315	m/s
FilterState_Max_NofObj	32	12	0	4095	1	
FilterState_Max_Distance	32	12	0	409.5	0.1	m
FilterState_Max_Azimuth	32	12	-50	52.375	0.025	deg
FilterState_Max_VrelOncome	32	12	0	128.993	0.0315	m/s
FilterState_Max_VrelDepart	32	12	0	128.993	0.0315	m/s
FilterState_Max_RCS	32	12	-50	52.375	0.025	dBm ²
FilterState_Max_Lifetime	32	12	0	409.5	0.1	s
FilterState_Max_Size	32	12	0	102.375	0.025	m ²
FilterState_Max_ProbExists	32	12	0	7	1	0x0: 0% 0x1: 25% 0x2: 50% 0x3: 75% 0x4: 90% 0x5: 99% 0x6: 99.9% 0x7: 100%
FilterState_Max_Y	32	12	-409.5	409.5	0.2	m
FilterState_Max_X	32	13	-500	1138.2	0.2	m
FilterState_Max_VYLeftRight	32	12	0	128.993	0.0315	m/s
FilterState_Max_VXOncome	32	12	0	128.993	0.0315	m/s
FilterState_Max_VYRightLeft	32	12	0	128.993	0.0315	m/s
FilterState_Max_VXDepart	32	12	0	128.993	0.0315	m/s

Table 17: FilterState_Cfg - signal description (0x204).

Start	Signal	Description
2	FilterState_Active	Filter activation switch
3	FilterState_Index	Multiplexor to specify which filter criterion to configure (0x0 to 0xF)
7	FilterState_Type	Multiplexor to choose between object or cluster filter configuration
16	FilterState_Min_xxx	Minimum value for filter criterion. The content depends on FilterState_Index (see Table 15 for more details)
32	FilterState_Max_xxx	Maximum value for filter criterion. The content depends on FilterState_Index (see Table 15 for more details)

7.4. Collision detection state (0x408)

Region-based collision detection can be activated with message **CollDetCfg** (0x400). When collision detection is active, the sensor sends the current collision detection configuration and warning state in message **CollDetState** (0x408) and for the individual regions in messages **CollDetRegionState** (0x402) cyclically (once per second).

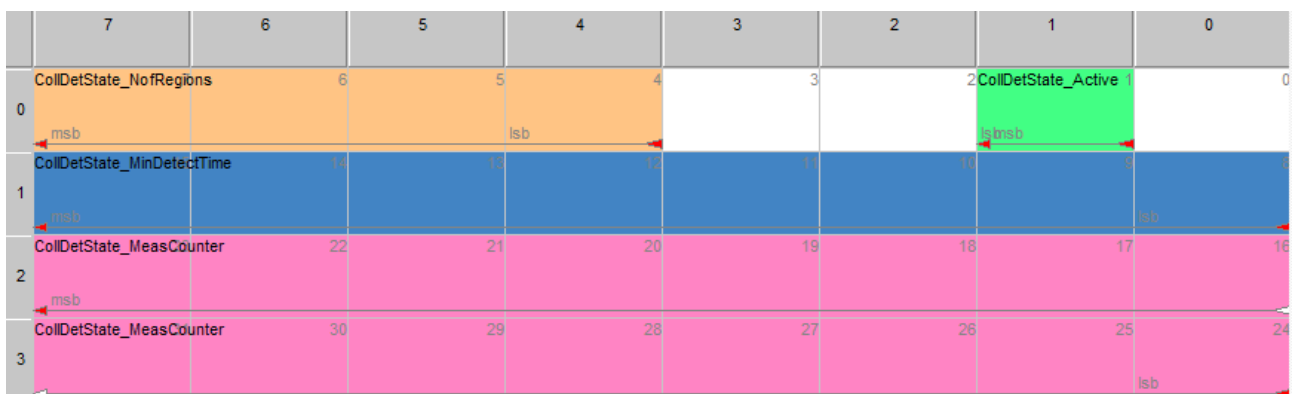


Figure 13: CollDetState - message layout (0x408).

Table 18: CollDetState - message content (0x408).

Signal	Start	Len	Min	Max	Res	Unit
CollDetState_Activation	1	1	0	1	1	0x0: inactive 0x1: active
CollDetState_NofRegions	4	4	0	8	1	-
CollDetState_MinDetectTime	8	8	0.0	25.5	0.1	sec
CollDetState_MeasCounter ⁷	24	16	0	65535	1	-

⁷ Not applicable for SRR308, signal value is to be ignored.

Table 19: CollDetState - signal description (0x408).

Start	Signal	Description
1	CollDetState_Activation	State of collision detection activation
4	CollDetState_NofRegions	Number of configured regions
8	CollDetState_MinDetectTime	Current configuration for minimum detection time of objects before a warning is triggered
24	CollDetState_MeasCounter	Measurement cycle counter (counting up since startup of sensor and restarting at 0 when > 65535)

7.5. Collision detection region state (0x402)

When collision detection is active, the sensor sends the current collision detection configuration and warning state in message **CollDetState** (0x408) and for the individual regions in messages **CollDetRegionState** (0x402) cyclically (once per second).

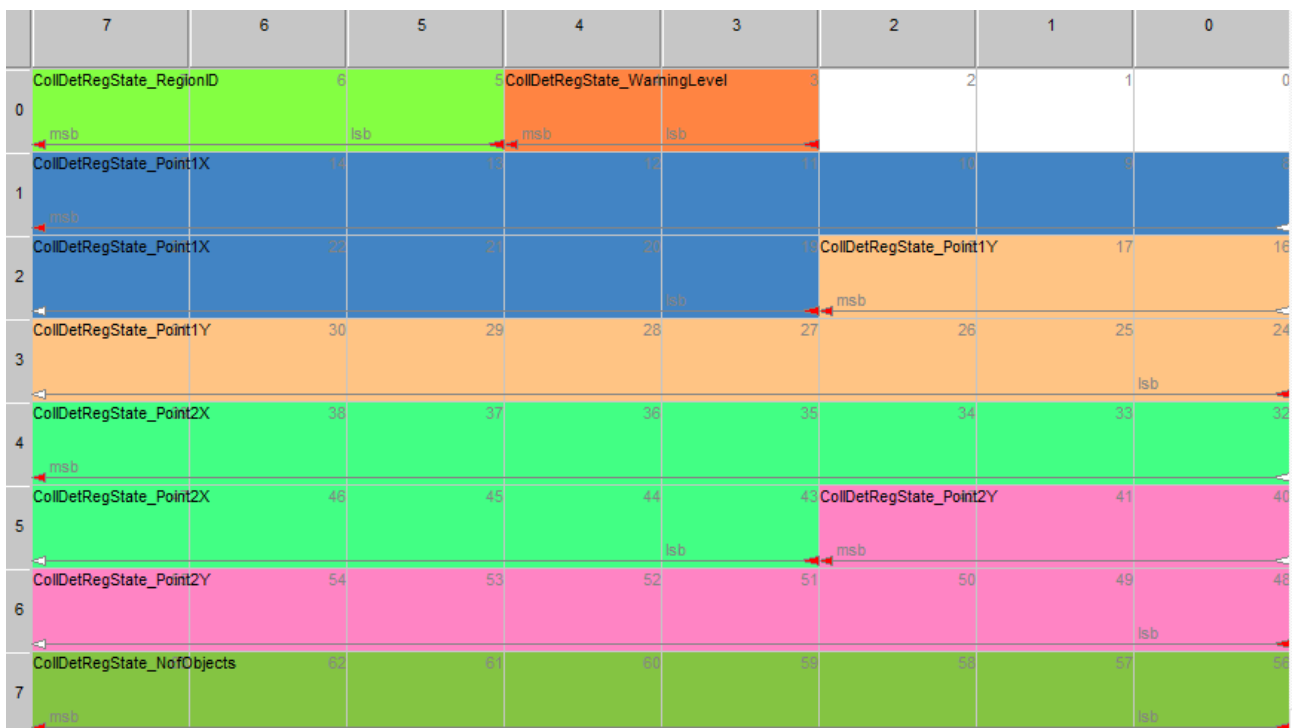


Figure 14: CollDetRegionState - message layout (0x402).

Table 20: CollDetRegionState - message content (0x402).

Signal	Start	Len	Min	Max	Res	Unit
CollDetRegState_WarningLevel	3	2	0	0	1	0x0: no warning 0x1: object warning 0x2: unused 0x3: warning deactivated
CollDetRegState_RegionID	5	3	0	7	1	-
CollDetRegState_Point1X	19	13	-500	1138.2	0.2	m
CollDetRegState_Point1Y	24	11	-204.6	204.8	0.2	m
CollDetRegState_Point2X	43	13	-500	1138.2	0.2	m
CollDetRegState_Point2Y	48	11	-204.6	204.8	0.2	m
CollDetRegState_NofObjects	56	8	0	255	1	-

Table 21: CollDetRegionState - signal description (0x402).

Start	Signal	Description
3	CollDetRegState_WarningLevel	Indicates that an object is inside the region or that it has ever been inside the region. If it is 0x0, no object has ever been detected inside the region. If it is 0x1, an object is currently inside the region and has exceeded the minimum detection time configured in CollDetCfg_MinTime. 0x2 keeps unused If it is 0x3, an object has been inside this region in the past, but has already left. To reset the warning level to 0x0, use CollDetCfg_WarningReset (see also section 6.3).
5	CollDetRegState_RegionID	ID of current region
19	CollDetRegState_Point1X	1 st X (longitudinal) coordinate of rectangular region
24	CollDetRegState_Point1Y	1 st Y (lateral) coordinate of rectangular region
43	CollDetRegState_Point2X	2 nd X (longitudinal) coordinate of rectangular region
48	CollDetRegState_Point2Y	2 nd Y (lateral) coordinate of rectangular region
56	CollDetRegState_NofObjects	Number of objects currently detected inside the region that fulfil the conditions for collision detection.

7.6. Control message for relay (0x8)

The ARS40X Collision detection supports a Wilke CAN Bus Relay for direct switching of external hardware. When collision detection is active and the relay control output is activated with **RadarCfg_CtrlRelay**, the sensor sends the relay control message **CollDetRelayCtrl** (0x8) cyclically (once per second).

Four different relays (i.e. the first four collision regions) are supported. The module has to be terminated on the CAN bus on both ends with a 120 Ohm resistor.



Figure 15 CAN-Bus I/O Module: 4 digital outputs (Article Code DV-CANFRAS4-01)

To use this module it has to be configured to 500kBit/s. The module address has to be configured to 0x8 (Upper switch: 0, bottom switch: 1).

The sensors send the relay control message all with the same message ID (independently on the chosen sensor ID). Thus, only one sensor in a network of sensor should be sending this message. As response, the CAN Bus Relay sends out a cyclic status message as response with CAN ID 0x10 when it is working.

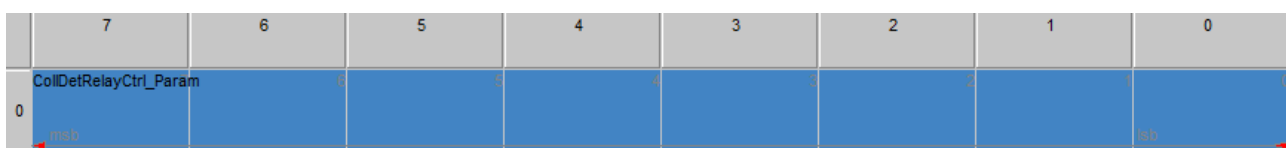


Figure 16: CollDetRelayCtrl - message layout (0x8).

Table 22: CollDetRelayCtrl - message content (0x8).

Signal	Start	Len	Min	Max	Res	Unit
CollDetRelayCtrl_Param	0	8	0	255	1	Bit 0: Relay 1 Bit 1: Relay 2 Bit 2: Relay 3 Bit 3: Relay 4 Bit 4: Reserved Bit 5: Reserved Bit 6: Reserved Bit 7: Reserved

Table 23: CollDetRelayCtrl - signal description (0x8).

Start	Signal	Description
0	CollDetRelayCtrl_Param	Control bit field for the 4 Relays on the module

7.7. Software Version Identification (0x700)

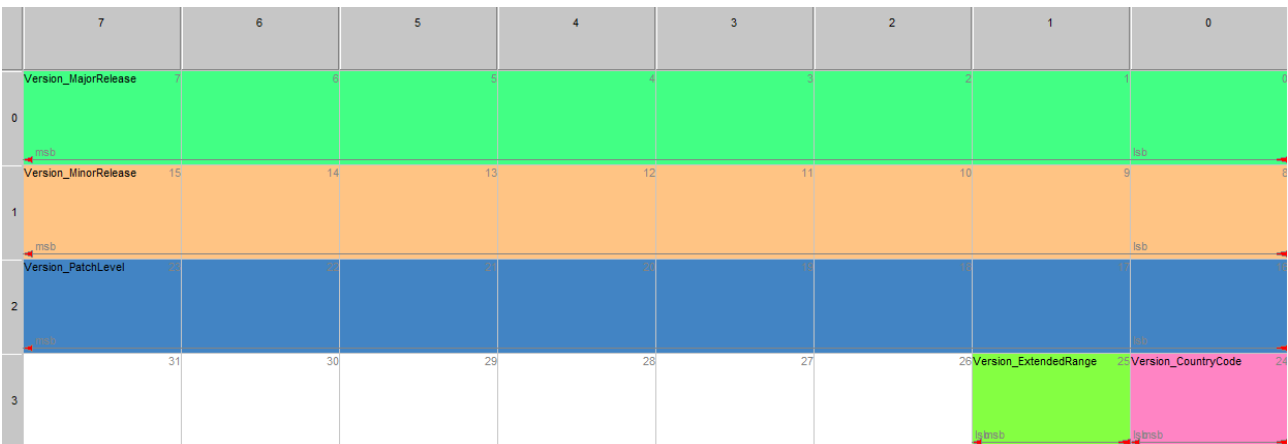


Figure 17: VersionID - message layout (0x700).

Table 24: VersionID - message content (0x700).

Signal	Start	Len	Min	Max	Res	Unit
Version_MajorRelease	0	8	0	256	1	
Version_MinorRelease	8	8	0	256	1	
Version_PatchLevel	16	8	0	256	1	

Version_ExtendedRange	25	1	0	1	1	0x0 – Standard Range Version 0x1 – Extended Range Version without any increase of the Power of Radiation
Version_CountryCode	24	1	0	1	1	0x0 – International Version 0x1 – South Korea, Japan – Power Reduction is demanded

Table 25: VersionID – signal description (0x700).

Start	Signal	Description
0	Version_MajorRelease	Software Main Version
8	Version_MinorRelease	Software Minor Version
16	Version_PatchLevel	Software Patch Level
24	Version_CountryCode ⁸	The version for South Korea / Japan has reduced power output of -3dB.
25	Version_ExtendedRange	The maximum range can be configured with the following limits. ARS408: Standard Range Version: 196 – 260 m Extended Range Version: 196 – 1200 m ARS404: Standard Range Version: 150 – 190 m Extended Range Version: 90 – 1000 m SRR308: Standard Range Version: 96 m Extended Range Version: 96 - 192 m

⁸ Not applicable in case of SRR308 and this signal value is to be ignored.

8. Motion Input Signals

The sensor accepts input messages 0x300 and 0x301, but will still function without these two messages. After a 500 ms timeout, the sensor will default to the following states:

1. **Speed** 0 m/s and standstill
2. **Yaw rate** 0 deg/s

Each message has independent timeout monitoring. The timeout state is reflected in signal **RadarState_MotionRxState** (0x201).

The input signals are used to evaluate a vehicle course with is used to determine the movement of detected clusters and objects.

8.1. SpeedInformation (0x300)

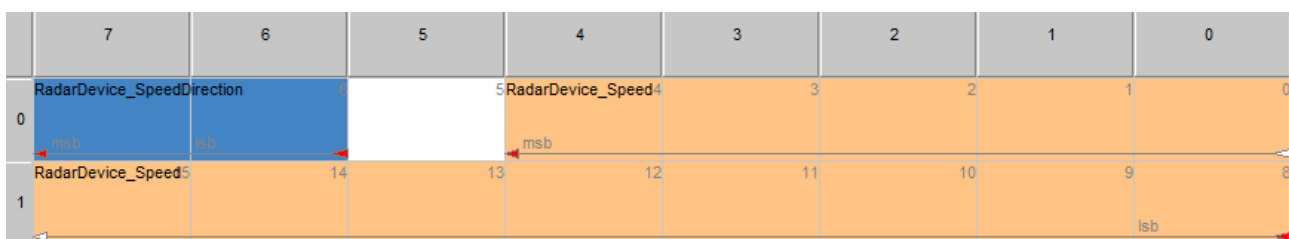


Figure 18: SpeedInformation - message layout (0x300).

Table 26: SpeedInformation - message content (0x300).

Signal	Start	Len	Min	Max	Res	Unit
RadarDevice_SpeedDirection	6	2	0	2	1	0x0: standstill 0x1: forward 0x2: backward
RadarDevice_Speed	8	13	0	163.8	0.02	m/s

Table 27: SpeedInformation – signal description (0x300).

Start	Signal	Description
6	RadarDevice_SpeedDirection	Indicates the direction of the radar movement while looking into positive straight ahead direction
8	RadarDevice_Speed	Absolute magnitude of speed in the direction the radar is moved while looking into positive straight ahead direction

8.2. Yaw rate Information (0x301)

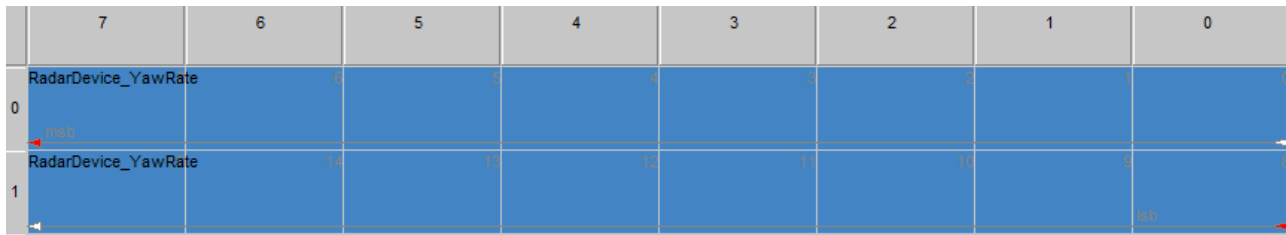


Figure 19: YawRateInformation - message layout (0x301).

Table 28: YawRateInformation - message content (0x301).

Signal	Start	Len	Min	Max	Res	Unit
RadarDevice_YawRate	8	16	-327.68	327.68	0.01	deg/s

Table 29: YawRateInformation – signal description (0x301).

Start	Signal	Description
8	RadarDevice_YawRate	Rate of change of angular velocity looking into positive straight ahead direction. The center of rotation is assumed to be 1.95 m behind the sensor.

9. Cluster List

The Cluster list output consists of up to three message definitions that are sent in a regular interval (about 70 to 80 ms) if clusters are selected in signal **RadarCfg_OutputType** (0x200).

1. **Cluster_0_Status** (0x600) – The first message contains list header information, i.e. the number of near scan clusters and number of far scan clusters that are sent afterwards.
2. **Cluster_1_General** (0x701) – This message contains the position and velocity of the clusters and is sent repeatedly for all the detected clusters (first near scan, then far scan). Each of the two cluster lists is range sorted. If there are more than 250 clusters, only the first 250 clusters are sent.
3. **Cluster_2_Quality** (0x702) – This message contains the quality information of the clusters and is only sent if it was activated in signal **RadarCfg_SendQuality** (0x200). It is sent repeatedly for all clusters in the same way as message **Cluster_1_General** (0x701).

If the quality information is sent, first all messages of type **Cluster_1_General** (0x701) are sent and afterwards all messages of type **Cluster_2_Quality** (0x702).

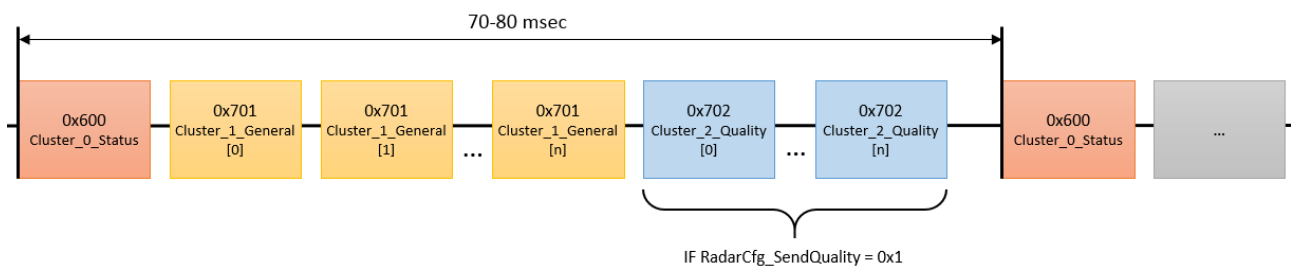


Figure 20: Overview of cluster list messages that are sent cyclically by the radar.

9.1. Cluster list status (0x600)

The message **Cluster_0_Status** (0x600) contains the cluster list header information and is sent as first message of the cluster list output and only once per measurement cycle.

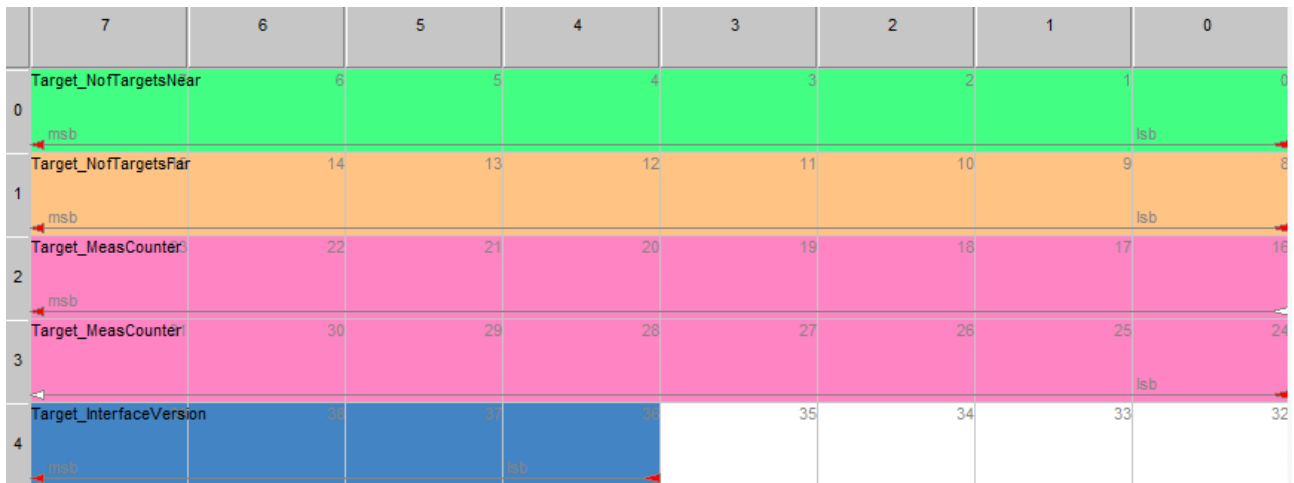


Figure 21: Cluster_0_Status - message layout (0x600).

Table 30: Cluster_0_Status - message content (0x600).

Signal	Start	Len	Min	Max	Res	Unit
Cluster_NofClustersNear	0	8	0	255	1	-
Cluster_NofClustersFar	8	8	0	255	1	-
Cluster_MeasCounter	24	16	0	65535	1	-
Cluster_InterfaceVersion	36	4	0	15	1	-

Table 31: Cluster_0_Status – signal description (0x600).

Start	Signal	Description
0	Cluster_NofClustersNear	Number of detected clusters in the near range scan (see below)
8	Cluster_NofClustersFar ⁹	Number of detected clusters in the far range scan (Near range and far range together max. 250 clusters; in case of activation of the quality mode app. 110 – 120 clusters before the CAN-Bus gets an overrun, which must be avoided)
24	Cluster_MeasCounter	Measurement cycle counter (counting up since startup of sensor and restarting at 0 when > 65535)
36	Cluster_InterfaceVersion	Cluster list CAN interface version

⁹ Only valid for ARS408

9.2. Cluster general information (0x701)

This message contains the position and velocity of the clusters and is sent repeatedly for all the detected clusters (first near scan, then far scan). Each of the two cluster lists is range sorted. If there are more than 250 clusters, only the first 250 clusters are sent.

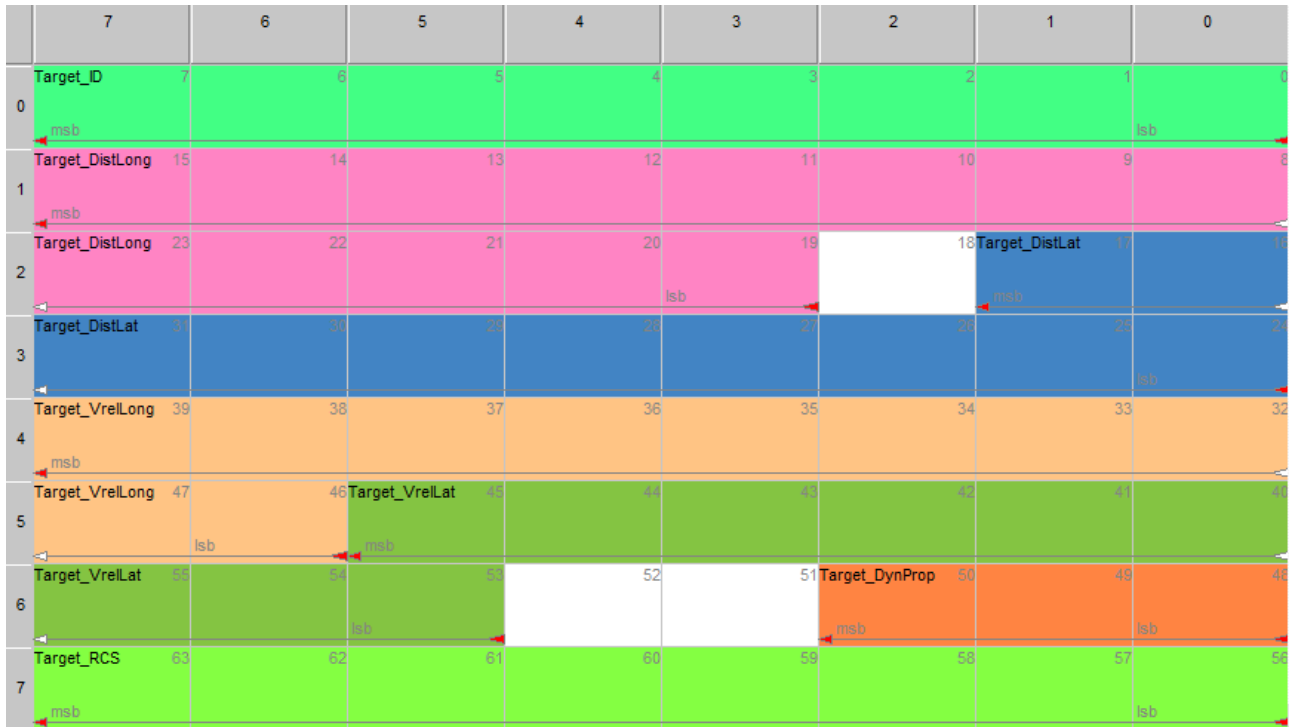


Figure 22: Cluster_1_General - message layout (0x701).

Table 32: Cluster_1_General - message content (0x701).

Signal	Start	Len	Min	Max	Res	Unit
Cluster_ID	0	8	0	255	1	
Cluster_DistLong	19	13	-500	+1138.2	0.2	m
Cluster_DistLat	24	10	-102.3	+102.3	0.2	m
Cluster_VrelLong	46	10	-128.00	127.75	0.25	m/s
Cluster_DynProp ¹⁰	48	3	0	7	1	0x0: moving 0x1: stationary 0x2: oncoming 0x3: stationary candidate 0x4: unknown 0x5: crossing stationary 0x6: crossing moving 0x7: stopped

¹⁰ Not applicable in case of SRR308 and this signal value is to be ignored.

Cluster_VrelLat	53	9	-64.00	63.75	0.25	m/s
Cluster_RCS	56	8	-64.0	63.5	0.5	dBm ²

Table 33: Cluster_1_General - signal description (0x701).

Start	Signal	Description
0	Cluster_ID	Cluster number
19	Cluster_DistLong	Longitudinal (x) coordinate
24	Cluster_DistLat	Lateral (y) coordinate
46	Cluster_VrelLong	Relative velocity in longitudinal direction (x) $V_{relLong} = \frac{V_{relDoppler}}{\cos \alpha}$ $V_{relDoppler} = \text{Relative velocity measured by the radar in radial direction}$ $\alpha = \tan^{-1}\left(\frac{Cluster_{DistLat}}{Cluster_{DistLong}}\right)$
53	Cluster_VrelLat	Relative velocity in lateral direction (y) This value is considered to be always 0.
48	Cluster_DynProp ¹¹	Dynamic property of cluster to indicate if it is moving or not
56	Cluster_RCS	Radar cross section

9.3. Cluster quality information (0x702)¹²

This message contains the quality information of the clusters and is only sent if it was activated in signal **RadarCfg_SendQuality** (0x200). It is sent repeatedly for all clusters in the same way as message **Cluster_1_General** (0x701). If this message is activated the maximum number of clusters has to be limited to 125 either by filtering or environment.

¹¹ Not applicable in case of SRR308 and this signal value is to be ignored.

¹² For SRR308, the Cluster quality information (0x702) message differs, please see section 9.4 for details.

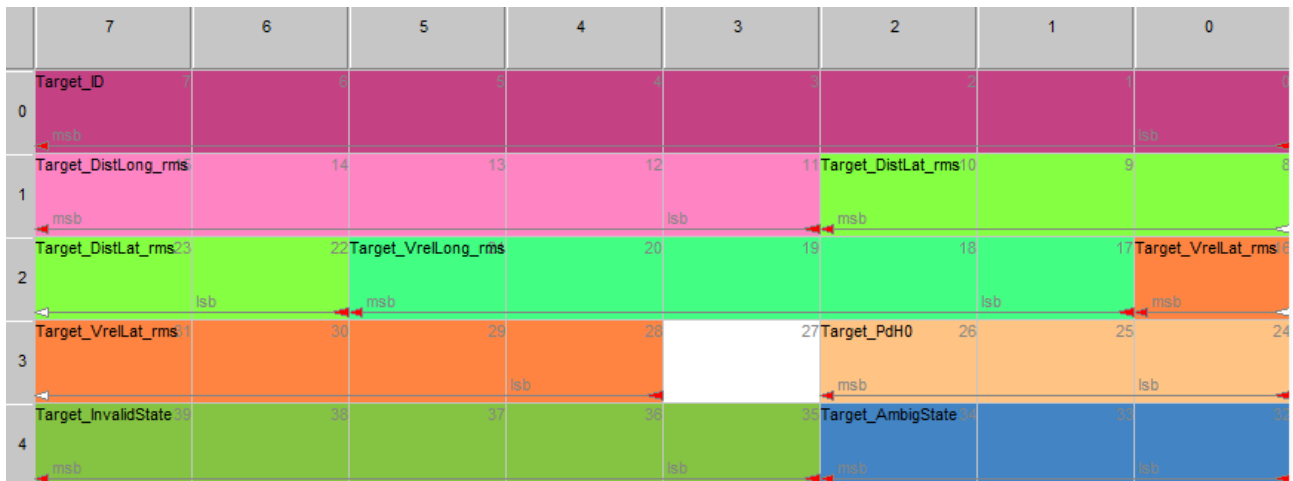


Figure 23: Cluster_2_Quality - message layout (0x702).

Table 34: Cluster_2_Quality - message content (0x702).

Signal	Start	Len	Min	Max	Res	Unit
Cluster_ID	0	8	0	255	1	-
Cluster_DistLong_rms	11	5	0	31	1	see Table 36
Cluster_VrelLong_rms	17	5	0	31	1	see Table 36
Cluster_DistLat_rms	22	5	0	31	1	see Table 36
Cluster_Pdh0	24	3	0	7	1	0x0: invalid 0x1: <25% 0x2: <50% 0x3: <75% 0x4: <90% 0x5: <99% 0x6: <99.9% 0x7: <=100%
Cluster_VrelLat_rms	28	5	0	31	1	see Table 36

Cluster_AmbigState	32	3	0	7	1	0x0: invalid (cannot be used) 0x1: ambiguous (cluster is not clear as there is fuzziness (darkness, ambiguity)). (Should not be used and not recommended anyway, because it could be there twice or even more times) 0x2: staggered ramp (the cluster is not clear somehow as the fuzziness (ambiguity) has been resolved partially). (Should not be used and not recommended anyway) 0x3: unambiguous (everything is clear, the fuzziness(ambiguity) has been resolved – it can be used and recommended for usage) 0x4: stationary candidates (everything is clear and probably there is something stationary (fixed) as the fuzziness(ambiguity) has been resolved – it can be used and recommended for usage)
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Cluster_InvalidState	35	5	0	31	1	0x00: Valid 0x01: Invalid due to low RCS 0x02: Invalid due to near-field artefact 0x03: Invalid far range Cluster because not confirmed in near range 0x04: Valid Cluster with low RCS 0x05: reserved 0x06: Invalid Cluster due to high mirror probability 0x07: Invalid because outside sensor field of view 0x08: Valid Cluster with azimuth correction due to elevation 0x09: Valid Cluster with high child probability 0x0A: Valid Cluster with high probability of being a 50 deg artefact 0x0B: Valid Cluster but no local maximum 0x0C: Valid Cluster with high artefact probability 0x0D: reserved 0x0E: Invalid Cluster because it is a harmonics 0x0F: Valid Cluster above 95 m in near range 0x10: Valid Cluster with high multi-target probability 0x11: Valid Cluster with suspicious angle
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Table 35: Cluster_2_Quality - signal description (0x702).

Start	Signal	Description
0	Cluster_ID	Cluster number
11	Cluster_DistLong_rms	Standard deviation of longitudinal distance
22	Cluster_DistLat_rms	Standard deviation of lateral distance
17	Cluster_VrelLong_rms	Standard deviation of longitudinal relative velocity
28	Cluster_VrelLat_rms	Standard deviation of lateral relative velocity

24	Cluster_Pdh0	False alarm probability of Cluster (i.e. probability for being an artefact caused by multipath or similar)
32	Cluster_AmbigState	State of Doppler (radial velocity) ambiguity solution (see the seventh row in table 34 for more details)
35	Cluster_InvalidState	State of Cluster validity state (see Table 34 for more details)

Table 36: Signal value table for Cluster_DistLong_rms, Cluster_DistLat_rms, Cluster_VrelLong_rms and Cluster_VrelLat_rms (0x702).

Parameter	Value for signals Cluster_DistLong_rms, Cluster_DistLat_rms [m] Cluster_VrelLong_rms, Cluster_VrelLat_rms [m/s]
0x0	<0.005
0x1	<0.006
0x2	<0.008
0x3	<0.011
0x4	<0.014
0x5	<0.018
0x6	<0.023
0x7	<0.029
0x8	<0.038
0x9	<0.049
0xA	<0.063
0xB	<0.081
0xC	<0.105
0xD	<0.135
0xE	<0.174
0xF	<0.224
0x10	<0.288
0x11	<0.371
0x12	<0.478
0x13	<0.616
0x14	<0.794
0x15	<1.023
0x16	<1.317
0x17	<1.697
0x18	<2.187
0x19	<2.817
0x1A	<3.630

0x1B	<4.676
0x1C	<6.025
0x1D	<7.762
0x1E	<10.000
0x1F	invalid

9.4. Cluster quality information (0x702) - SRR308

This message contains the quality information of the clusters and is only sent if it was activated in signal **RadarCfg_SendQuality** (0x200). It is sent repeatedly for all clusters in the same way as message **Cluster_1_General** (0x701). If this message is activated the maximum number of clusters has to be limited to 70 either by filtering or environment.

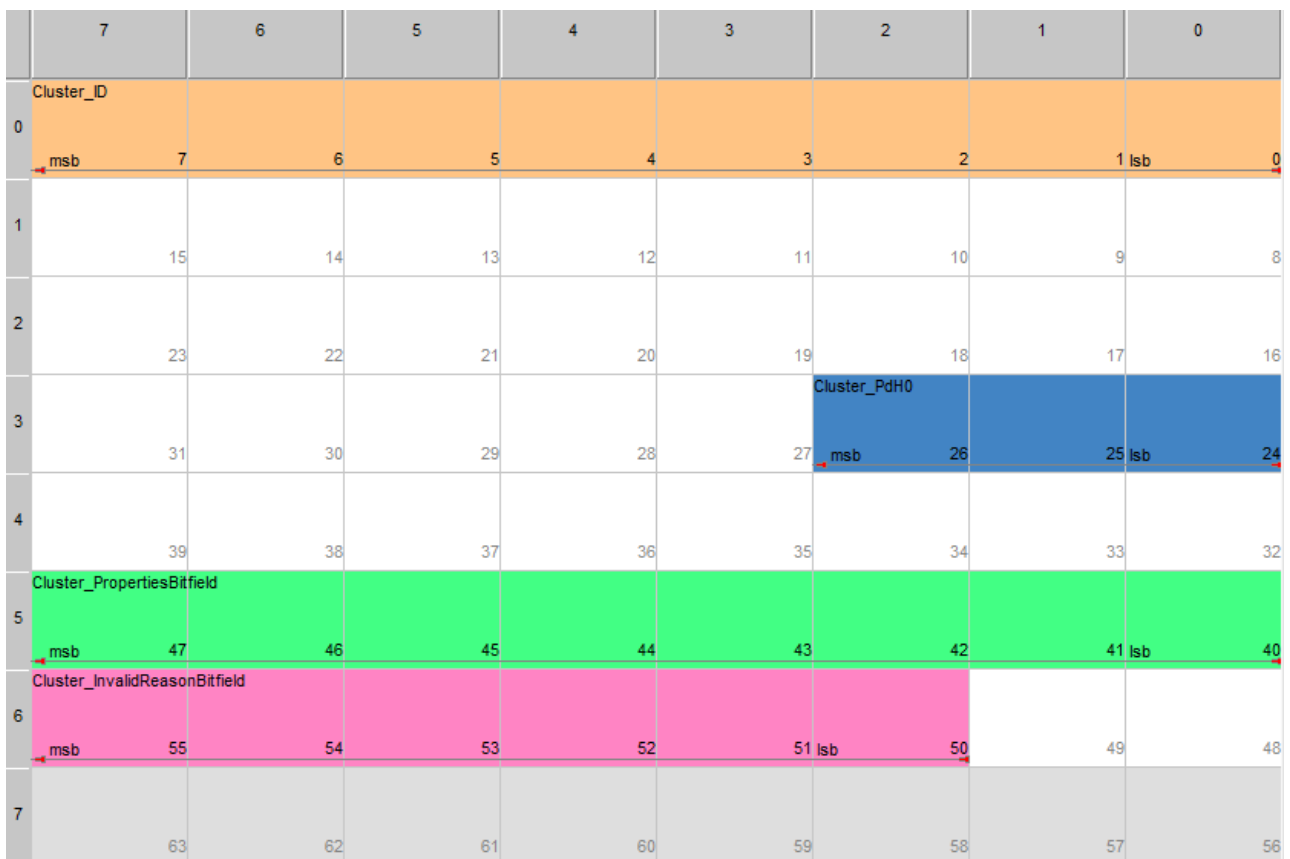


Figure 24: Cluster_2_Quality - message layout (0x702) – SRR308.

Table 37: Cluster_2_Quality - message content (0x702) – SRR308.

Signal	Start	Len	Min	Max	Res	Unit
Cluster_ID	0	8	0	255	1	-

Cluster_Pdh0	24	3	0	7	1	0x0: invalid 0x1: <25% 0x2: <50% 0x3: <75% 0x4: <90% 0x5: <99% 0x6: <99.9% 0x7: <=100%
Cluster_PropertiesBitfield	40	8	0	255	1	-
Cluster_InvalidReasonBitfield	50	6	0	63	1	-

Table 38: Cluster_2_Quality - signal description (0x702) – SRR308.

Start	Signal	Description
0	Cluster_ID	Cluster number
24	Cluster_Pdh0	False alarm probability of Cluster (i.e. probability for being an artefact caused by multipath or similar)
40	Cluster_PropertiesBitfield	Bit field indicating various properties of Cluster 0x0: Undefined property Bit 0: Static Cluster Bit 1: Dynamic Cluster Bit 2: Ambiguous Cluster Bit 4: Cluster from near scan Bit 6: Cluster behind guard rail
50	Cluster_InvalidReasonBitfield	Bit field indicating reason for invalid cluster 0x0: Valid Cluster Bit 1: Dynamic Cluster with low RCS Bit 2: Static Cluster with low RCS Bit 3: Cluster range rate improbable Bit 4: Cluster range improbable Bit 5: Cluster is a reflection of Ego

10. Object List

The object list output consists of up to five message definitions that are sent in a regular interval if objects are selected in signal **RadarCfg_OutputType** (0x200).

1. **Object_0_Status** (0x60A) – The first message contains list header information, i.e. the number of objects that are sent afterwards
2. **Object_1_General** (0x60B) – This message contains the position and velocity of the objects and is sent repeatedly for all the tracked objects.
3. **Object_2_Quality** (0x60C) – This message contains the quality information of the objects and is only sent if it was activated in signal **RadarCfg_SendQuality**

(0x200). It is sent repeatedly for all objects in the same way as message **Object_1_General** (0x60B).

4. **Object_3_Extended** (0x60D) – This message contains additional object properties and is only sent if it was activated in signal **RadarCfg_SendExtInfo** (0x200). It is sent repeatedly for all objects in the same way as message **Object_1_General** (0x60B).
5. **Object_4_Warning** (0x60E). This message contains the collision detection warning state and is only sent if collision detection was activated in message **CollDetCfg** (0x400). It is sent repeatedly for all objects in the same way as message **Object_1_General** (0x60B).

If the quality information, extended information and/or warning state is sent, first all messages of type **Object_1_General** (0x60B) are sent and afterwards all messages of type **Object_2_Quality** (0x60C), afterwards of type **Object_3_Extended** (0x60D) and/or afterwards of type **Object_4_Warning** (0x60E).

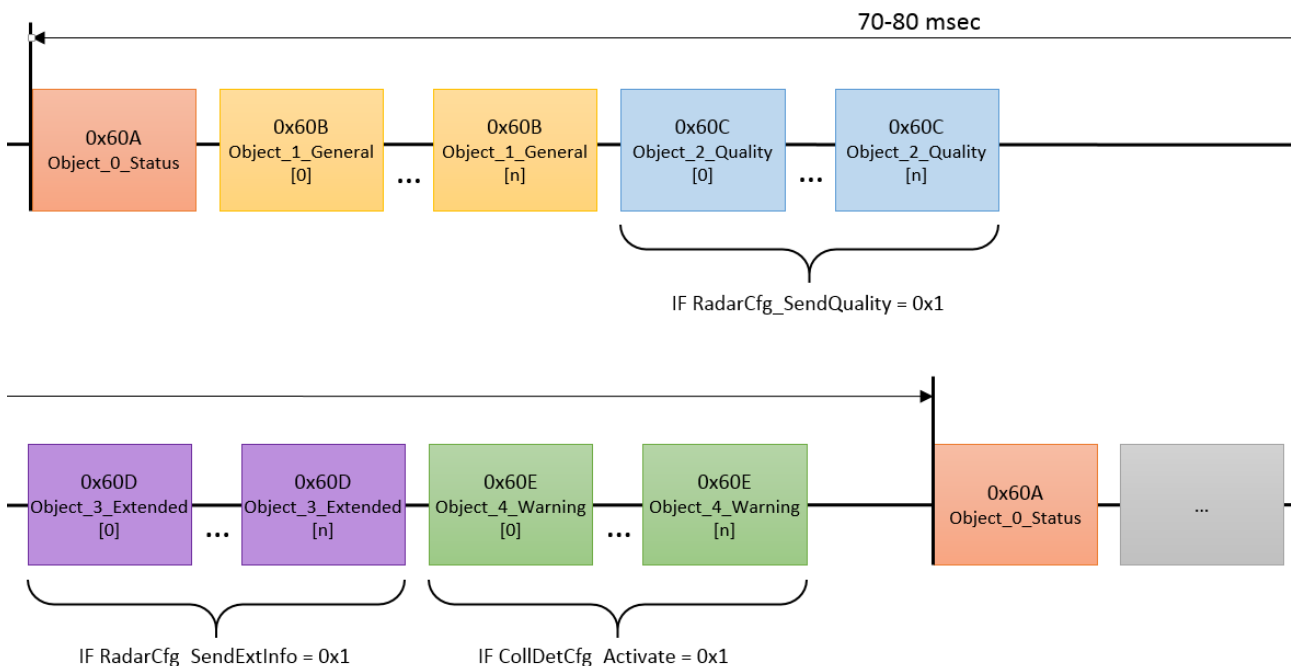


Figure 25: Overview of object list messages that are sent cyclically by the radar.

10.1. Object list status (0x60A)

The message **Object_0_Status** (0x60A) contains the object list header information and is sent as first message of the cluster list output and only once per measurement cycle.

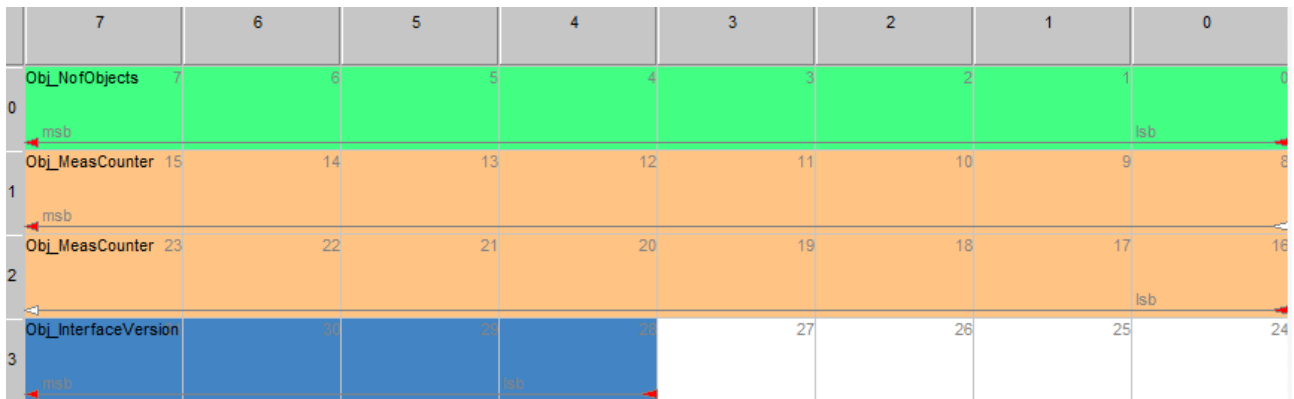


Figure 26: Object_0_Status - message layout (0x60A).

Table 39: Object_0_Status - message content (0x60A).

Signal	Start	Len	Min	Max	Res	Unit
Object_NofObjects	0	8	0	255	1	-
Object_MeasCounter	16	16	0	65535	1	-
Object_InterfaceVersion	28	4	0	15	1	-

Table 40: Object_0_Status - signal description (0x60A)

Start	Signal	Description
0	Object_NofObjects	Number of objects (max. 100 Objects)
16	Object_MeasCounter	Measurement cycle counter (counting up since startup of sensor and restarting at 0 when > 65535)
28	Object_InterfaceVersion	Object list CAN interface version

10.2. Object general information (0x60B)

This message contains the position and velocity of the objects and is sent repeatedly for all the tracked objects.



Figure 27: Object_1_General - message layout (0x60B).

Table 41: Object_1_General - message content (0x60B).

Signal	Start	Len	Min	Max	Res	Unit
Object_ID	0	8	0	255	1	
Object_DistLong	19	13	-500	+1138.2	0.2	m
Object_DistLat	24	11	-204.6	+204.8	0.2	m
Object_VrelLong	46	10	-128.00	127.75	0.25	m/s
Object_DynProp	48	3	0	7	1	0x0: moving 0x1: stationary 0x2: oncoming 0x3: crossing left 0x4: crossing right 0x5: unknown 0x6: stopped
Object_VrelLat	53	9	-64.00	63.75	0.25	m/s
Object_RCS	56	8	-64.0	63.5	0.5	dBm ²

Table 42: Object_1_General - signal description (0x60B).

Start	Signal	Description
0	Object_ID	Object ID (since objects are tracked, the ID is kept throughout measurement cycles and does not have to be consecutive)
19	Object_DistLong	Longitudinal (x) coordinate
24	Object_DistLat	Lateral (y) coordinate
46	Object_VrelLong	Relative velocity in longitudinal direction (x)
53	Object_VrelLat	Relative velocity in lateral direction (y)
48	Object_DynProp	Dynamic property of the object indicating if the object is moving or stationary (this value can only be determined correctly if the speed and yaw rate is given correctly)
56	Object_RCS	Radar cross section

10.3. Object quality information (0x60C)

This message contains the quality information of the objects and is only sent if it was activated in signal **RadarCfg_SendQuality** (0x200). It is sent repeatedly for all objects in the same way as message **Object_1_General** (0x60B).

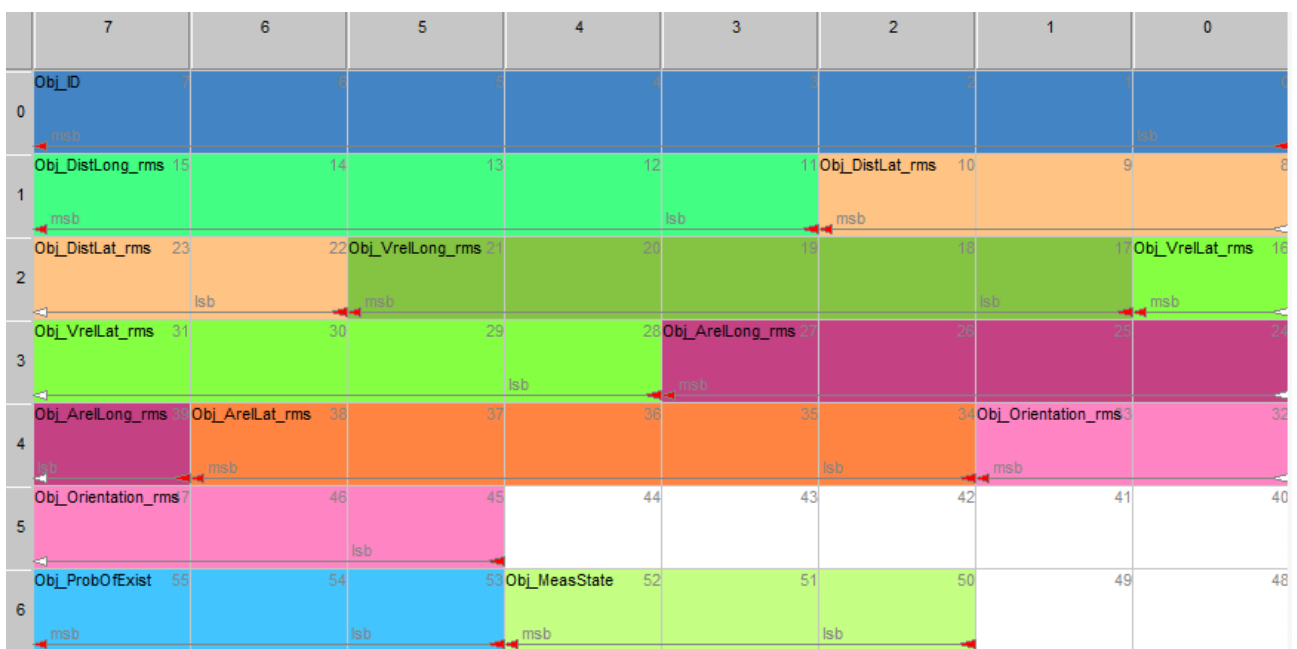


Figure 28: Object_2_Quality - message layout (0x60C).

Table 43: Object_2_Quality - message content (0x60C).

Signal	Start	Len	Min	Max	Res	Unit
Obj_ID	0	8	0	255	1	
Obj_DistLong_rms	11	5	0	31	1	see Table 45
Obj_VrelLong_rms	17	5	0	31	1	see Table 45
Obj_DistLat_rms	22	5	0	31	1	see Table 45
Obj_VrelLat_rms	28	5	0	31	1	see Table 45
Obj_ArelLat_rms	34	5	0	31	1	see Table 45
Obj_ArelLong_rms	39	5	0	31	1	see Table 45
Obj_Orientation_rms	45	5	0	31	1	deg
Obj_MeasState	50	3	0	7	1	0x0: deleted 0x1: new 0x2: measured 0x3: predicted 0x4: deleted for merge 0x5: new from merge
Obj_ProbOfExist	53	3	0	7	1	0x0: invalid 0x1: <25% 0x2: <50% 0x3: <75% 0x4: <90% 0x5: <99% 0x6: <99.9% 0x7: <=100%

Table 44: Object_2_Quality – signal description (0x60C).

Start	Signal	Description
0	Obj_ID	Object ID (since objects are tracked, the ID is kept throughout measurement cycles and does not have to be consecutive)
11	Obj_DistLong_rms	Standard deviation of longitudinal distance
17	Obj_VrelLong_rms	Standard deviation of longitudinal relative velocity
22	Obj_DistLat_rms	Standard deviation of lateral distance
28	Obj_VrelLat_rms	Standard deviation of lateral relative velocity
34	Obj_ArelLat_rms	Standard deviation of lateral relative acceleration
39	Obj_ArelLong_rms	Standard deviation of longitudinal relative acceleration
45	Obj_Orientation_rms	Standard deviation of orientation angle

50	Obj_MeasState	Measurement state indicating if the object is valid and has been confirmed by clusters in the new measurement cycle: 0x0: Deleted Object – Object has been deleted – is displayed during the last cycles of transmission just before the object ID disappears. 0x1: New Object is created – is displayed during the first cycles of transmission just after the object ID is created. Can also be checked by the character <i>Object_MeasCounter</i>. 0x2: Measured – Object creation has been confirmed by the actual measurement. Cluster could be created. 0x3: Predicted - Object creation could not be confirmed by the actual measurement. Cluster could not be created. 0x4: Deleted for merge - Object became deleted in order to be merged with another Object. 0x5: New from merge – new Object became created after a merge.
53	Obj_ProbOfExist	Probability of existence

Table 45: Signal value table for Obj_Orientation_rms, Obj_DistLong_rms, Obj_DistLat_rms, Obj_VrelLong_rms, Obj_VrelLat_rms, Obj_ArelLat_rms, Obj_ArelLong_rms (0x60C).

Parameter	Values for signal Obj_Orientation_rms [deg]	Value for signals Obj_DistLong_rms, Obj_DistLat_rms [m] Obj_VrelLong_rms, Obj_VrelLat_rms [m/s] Obj_ArelLat_rms, Obj_ArelLong_rms [m/s²]
0x0	<0.005	<0.005
0x1	<0.007	<0.006
0x2	<0.010	<0.008
0x3	<0.014	<0.011
0x4	<0.020	<0.014
0x5	<0.029	<0.018
0x6	<0.041	<0.023
0x7	<0.058	<0.029
0x8	<0.082	<0.038
0x9	<0.116	<0.049
0xA	<0.165	<0.063
0xB	<0.234	<0.081
0xC	<0.332	<0.105
0xD	<0.471	<0.135

0xE	<0.669	<0.174
0xF	<0.949	<0.224
0x10	<1.346	<0.288
0x11	<1.909	<0.371
0x12	<2.709	<0.478
0x13	<3.843	<0.616
0x14	<5.451	<0.794
0x15	<7.734	<1.023
0x16	<10.971	<1.317
0x17	<15.565	<1.697
0x18	<22.081	<2.187
0x19	<31.325	<2.817
0x1A	<44.439	<3.630
0x1B	<63.044	<4.676
0x1C	<89.437	<6.025
0x1D	<126.881	<7.762
0x1E	<180.000	<10.000
0x1F	invalid	invalid

10.4. Object extended information (0x60D)

This message contains additional object properties and is only sent if it was activated in signal **RadarCfg_SendExtInfo** (0x200). It is sent repeatedly for all objects in the same way as message **Object_1_General** (0x60B).

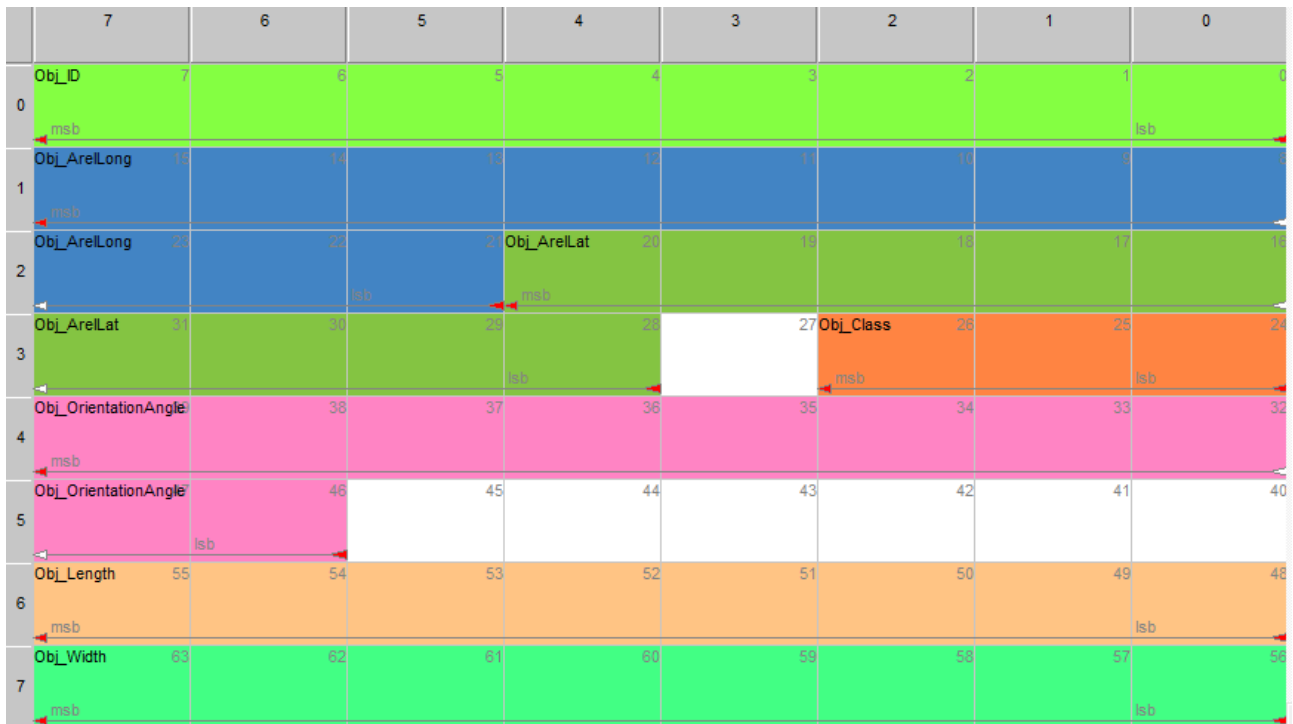


Figure 29: Object_3_Extended - message layout (0x60D).

Table 46: Object_3_Extended - message content (0x60D).

Signal	Start	Len	Min	Max	Res	Unit
Object_ID	0	8	0	255	1	
Object_AreLong	21	11	-10.00	10.47	0.01	m/s ²
Object_Class	24	3	0	7	1	0x0: point 0x1: car 0x2: truck 0x3: reserved 0x4: motorcycle 0x5: bicycle 0x6: wide 0x7: reserved
Object_AreLat	28	9	-2.50	2.61	0.01	m/s ²
Object_OrientationAngel	46	10	-180.00	180.00	0.4	deg
Object_Length	48	8	0.0	51.0	0.2	m
Object_Width	56	8	0.0	51.0	0.2	m

Table 47: Object_3_Extended - signal description (0x60D).

Start	Signal	Description
0	Object_ID	Object ID (since objects are tracked, the ID is kept throughout measurement cycles and does not have to be consecutive)

21	Object_ArelLong	Relative acceleration in longitudinal direction
24	Object_Class	0x0: point 0x1: car 0x2: truck 0x3: not used 0x4: motorcycle 0x5: bicycle 0x6: wide 0x7: reserved
28	Object_ArelLat	Relative acceleration in lateral direction
46	Object_OrientationAngel	Orientation angle of the object
48	Object_Length	Length of the tracked object
56	Object_Width	Width of the tracked object

10.5. Object collision detection warning (0x60E)

This message contains the collision detection warning state and is only sent if collision detection was activated in message **CollDetCfg** (0x400). It is sent repeatedly for all objects in the same way as message **Object_1_General** (0x60B).

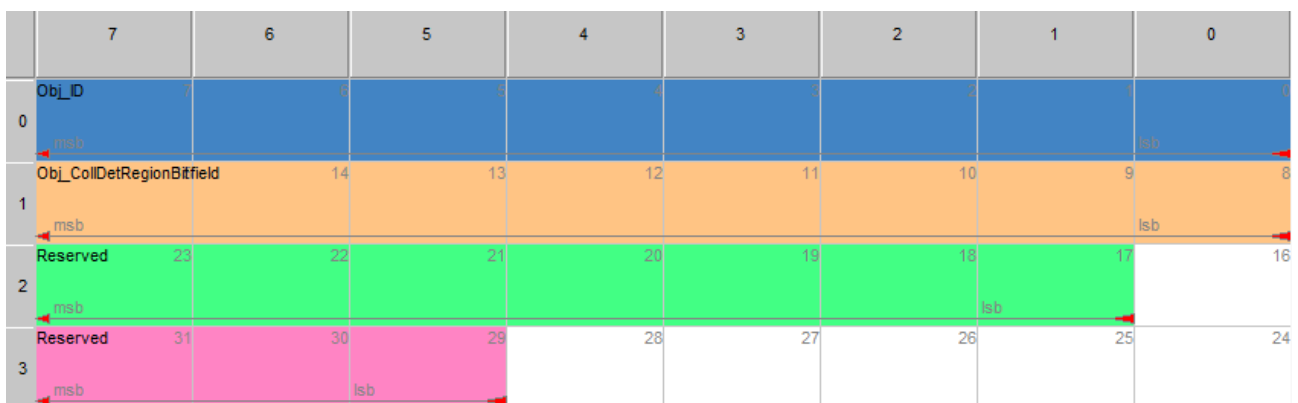


Figure 30: Object_4_Warning - message layout (0x60E).

Table 48: Object_4_Warning - message content (0x60E).

Signal	Start	Len	Min	Max	Res	Unit
Object_ID	0	8	0	255	1	
Object_CollDetRegionBitfield	8	8	0	255	1	

Table 49: Object_4_Warning - signal description (0x60E).

Start	Signal	Description
0	Object_ID	Object ID (since objects are tracked, the ID is kept throughout measurement cycles and does not have to be consecutive)
8	Object_CollDetRegionBitfield	Bit field of the regions, with bit set to 1 for regions that have a collision with this object